Solutions to the theoretical questions from ISLP

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Chapter 2: Statistical Learning

1. For each of parts (a) through (d), indicate whether we would generally expect the performance of a flexible statistical learning method to be better or worse than an inflexible method. Justify your answer.

   (a) **The sample size n is extremely large, and the number of predictors p is small.**

   > **Solution: Better.** When the sample size is very large and the number of predictors is small, a flexible method is expected to perform better. With ample data, flexible methods can accurately capture complex, potentially non-linear relationships between predictors and the response without overfitting to noise. The large n helps to mitigate the variance issues that often accompany flexible models.

   (b) **The number of predictors p is extremely large, and the number of observations n is small.**

   > **Solution: Worse.** In this scenario, flexible methods are highly likely to overfit the training data. With a small number of observations relative to a large number of predictors, a flexible model can perfectly fit the noise in the training data, leading to very high variance and consequently poor performance on unseen test data. Inflexible methods, by being more restrictive, can help control this variance.

   (c) **The relationship between the predictors and response is highly non-linear.**

   > **Solution: Better.** When the true relationship between the predictors and response is highly non-linear, a flexible method is better suited to capture this complex pattern. Inflexible methods, such as linear regression, would suffer from high bias and would not be able to accurately model the true functional form, leading to poor predictions.

   (d) **The variance of the error terms, i.e. $\sigma^2 = \text{Var}(\epsilon)$, is extremely high.**

   > **Solution: Worse.** When the error variance is very high, there is a lot of inherent "noise" in the data. Flexible methods tend to fit this noise very closely, leading to overfitting. This results in high variance and poor generalization to new observations. In such cases, a less flexible method that captures the underlying trend without chasing the noise would typically perform better.

2. Explain whether each scenario is a classification or regression problem, and indicate whether we are most interested in inference or prediction. Finally, provide n and p.

   (a) **We collect a set of data on the top 500 firms in the US. For each firm we record profit, number of employees, industry and the CEO salary. We are interested in understanding which factors affect CEO salary.**

   > **Solution:**
   >
   > - **Problem Type:** This is a **regression** problem because CEO salary is a quantitative (numerical) response variable.
   > - **Goal:** We are most interested in **inference** because the goal is to understand which factors affect CEO salary, implying a desire to interpret the relationships between predictors and the response.
   > - **n and p:**
   > - **n** (number of observations): 500 (number of firms)
   > - **p** (number of predictors): 3 (profit, number of employees, industry). CEO salary is the response variable.

(b) **We are considering launching a new product and wish to know whether it will be a success or a failure. We collect data on 20 similar products that were previously launched. For each product we have recorded whether it was a success or failure, price charged for the product, marketing budget, competition price, and ten other variables.**

Solution:

- **Problem Type:** This is a **classification** problem because product success or failure is a qualitative (categorical) response variable.
- **Goal:** We are most interested in **prediction** because the aim is to determine if a new product will be a success or failure.
- **n and p:**
  - **n** (number of observations): 20 (number of similar products)
  - **p** (number of predictors): 13 (price charged, marketing budget, competition price, and ten other variables)

(c) **We are interested in predicting the % change in the USD/Euro exchange rate in relation to the weekly changes in the world stock markets. Hence we collect weekly data for all of 2012. For each week we record the % change in the USD/Euro, the % change in the US market, the % change in the British market, and the % change in the German market.**

Solution:

- **Problem Type:** This is a **regression** problem because the % change in the USD/Euro exchange rate is a quantitative (numerical) response variable.
- **Goal:** We are most interested in **prediction** because the aim is to predict the % change in the exchange rate.
- **n and p:**
  - **n** (number of observations): 52 (weekly data for all of 2012)
  - **p** (number of predictors): 3 (% change in US market, % change in British market, % change in German market)

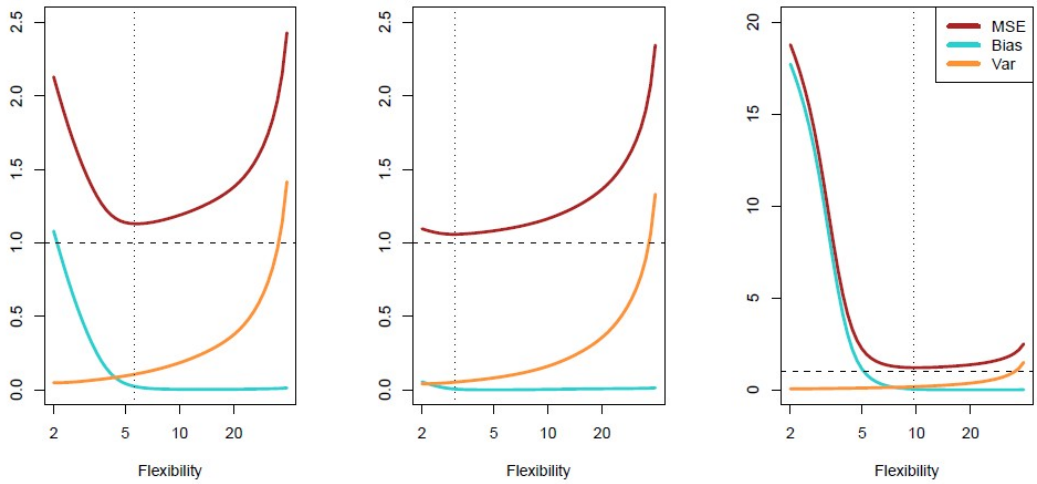
3. We now revisit the bias-variance decomposition.

(a) Provide a sketch of typical (squared) bias, variance, training error, test error, and Bayes (or irreducible) error curves, on a single plot, as we go from less flexible statistical learning methods towards more flexible approaches. The x-axis should represent the amount of flexibility in the method, and the y-axis should represent the values for each curve. There should be five curves. Make sure to label each one.

(b) Explain why each of the five curves has the shape displayed in part (a).

Solution:

(a) A sketch of the bias-variance decomposition curves is illustrated below (similar to Figure 2.12, in ISLP).



Caption: A sketch of the typical (squared) bias, variance, training error, test error, and Bayes (irreducible) error curves as model flexibility increases.

(b) **Explanation of the curves:**

- **Bayes (Irreducible) Error (Horizontal Dashed Line):** This curve is constant because the Bayes error represents the minimum possible test error rate achievable, even with a perfect model. It is the error that results from the inherent noise in the data itself, $\text{Var}(\epsilon)$, and cannot be reduced by any statistical learning method.
- **Squared Bias (Decreasing Curve):** Bias refers to the error introduced by approximating a real-world problem, which may be complicated, by a much simpler model. As the flexibility of a statistical learning method increases, its ability to capture the true underlying (potentially complex) relationship between predictors and the response improves. This leads to a decrease in bias.
- **Variance (Increasing Curve):** Variance refers to the amount by which the estimated function $\hat{f}$ would change if we estimated it using a different training data set. More flexible methods are highly sensitive to changes in the training data, meaning small changes can result in large changes in $\hat{f}$. Therefore, as flexibility increases, variance typically increases.
- **Training Error (Monotonically Decreasing Curve):** The training error continuously decreases as the model flexibility increases. A more flexible model can fit the training data more closely, including fitting the random errors or noise present in the training set. This leads to a lower (often approaching zero) error on the data it was trained on.
- **Test Error (U-shaped Curve):** The test error initially decreases as flexibility increases, benefiting from the reduction in bias. However, beyond a certain point, the test error begins to increase. This is because, after reaching the optimal balance, further increases in flexibility lead to a disproportionately large increase in variance (overfitting the noise) compared to any further reduction in bias. The model then performs poorly on new, unseen data. The expected test MSE is decomposed into the sum of variance, squared bias, and irreducible error.

4. You will now think of some real-life applications for statistical learning.

   (a) Describe three real-life applications in which classification might be useful. Describe the response, as well as the predictors. Is the goal of each application inference or prediction? Explain your answer.

   (b) Describe three real-life applications in which regression might be useful. Describe the response, as well as the predictors. Is the goal of each application inference or prediction? Explain your answer.

Solution:

(a) **Applications for Classification (qualitative response):**

i) **Application: Email Spam Detection**

- **Response:** Whether an email is "Spam" or "Not Spam" (binary, qualitative).
- **Predictors:** Content of the email (e.g., specific keywords, sender's address, presence of attachments, number of exclamation marks), metadata (e.g., time sent, recipient).
- **Goal: Prediction**. The primary goal is to accurately classify incoming emails so users don't have to deal with spam, rather than deeply understanding why specific words lead to spam.

ii) **Application: Customer Churn Prediction**

- **Response:** Whether a customer will "Churn" (cancel service) or "Not Churn" (binary, qualitative).
- **Predictors:** Customer demographics (age, income), usage patterns (call duration, data consumption), contract type, customer service interactions, previous complaints.
- **Goal: Prediction** and **Inference**. Companies want to predict which customers are likely to churn to intervene (prediction). They also want to understand *why* customers churn to improve services (inference).

iii) **Application: Medical Diagnosis of Disease**

- **Response:** Presence or absence of a specific disease (e.g., "Cancer" or "No Cancer") (binary, qualitative).
- **Predictors:** Patient symptoms (e.g., fever, pain level), medical test results (e.g., blood pressure, lab values), genetic markers, age, family history.
- **Goal: Inference** and **Prediction**. Doctors need to accurately predict if a patient has a disease. They also critically need to infer which symptoms and test results are most indicative of the disease to guide treatment and research.

(b) **Applications for Regression (quantitative response):**

i) **Application: House Price Prediction**

- **Response:** The sale price of a house (continuous, quantitative).
- **Predictors:** Property characteristics (e.g., square footage, number of bedrooms/bathrooms, lot size, age of house), location features (e.g., zip code, school district quality, proximity to amenities).
- **Goal: Prediction**. The main goal is to estimate the market value of a house as accurately as possible for buyers, sellers, or real estate agents.

ii) **Application: Advertising Budget Optimization**

- **Response:** Product sales (continuous, quantitative).
- **Predictors:** Advertising budgets for different media (e.g., TV, radio, newspaper).
- **Goal: Inference**. While predicting sales is possible, the core interest here is often to understand the impact of each advertising medium on sales, i.e., "which factors affect sales" and "how large is the association between each medium and sales" to inform future marketing strategy. This helps optimize budget allocation.

iii) **Application: Economic Growth Forecasting**

- **Response:** Gross Domestic Product (GDP) growth rate (continuous, quantitative).
- **Predictors:** Interest rates, inflation rates, unemployment figures, consumer spending, government investment.
- **Goal: Prediction**. Governments and financial institutions aim to forecast future economic growth to make informed policy and investment decisions. Understanding the individual impact of each predictor (inference) is also important for policy-making.

5. What are the advantages and disadvantages of a very flexible (versus a less flexible) approach for regression or classification? Under what circumstances might a more flexible approach be preferred to a less flexible approach? When might a less flexible approach be preferred?

Solution: The choice between a very flexible (non-parametric) approach and a less flexible (parametric) approach in regression or classification involves navigating the bias-variance trade-off. Less flexible methods, such as linear regression, impose strong assumptions about the functional form of the relationship, while more flexible methods, such as K-Nearest Neighbors (KNN) regression, do not explicitly assume a fixed functional form.

► **Advantages and Disadvantages of a Very Flexible (Non-Parametric) Approach:**

Advantages and Disadvantages of Flexible Approaches:

Advantage	**Disadvantage**
Low bias: Does not make strong assumptions about $f(X)$, allowing it to accurately model complex or non-linear structures.	**High variance:** The prediction is highly sensitive to individual training observations (especially with small K), leading to a less stable and highly variable fit.
Superior performance when the true relationship is strongly non-linear.	**Degrades quickly in higher dimensions:** Suffers from the "curse of dimensionality," meaning performance decreases rapidly as the number of predictors (p) increases, as observations become sparsely distributed.
Provides a means to perform regression when the true relationship is unknown.	May still perform worse than less flexible methods even in non-linear scenarios, particularly in high dimensions

Advantages and Disadvantages of Less Flexible Approaches:

Advantage	**Disadvantage**
Easy to fit: Requires estimating only a small number of coefficients.	**High bias:** Makes strong assumptions about the form of $f(X)$ (e.g., linearity and additivity).
Simple interpretation: Coefficients have straightforward interpretations (e.g., β_j quantifies the average effect on Y of a one-unit increase in X_j, holding others fixed).	**Poor performance:** If the specified linear form is far from the true relationship, prediction accuracy will be poor.
Statistical rigor: Tests of statistical significance (like p-values and confidence intervals) can be easily performed.	
Stability in high dimensions: Performance deteriorates slowly in the presence of additional noise variables (predictors not associated with the response) compared to flexible methods like KNN.	
Preferred when there is a small number of observations per predictor.	

► A **more flexible approach** might be preferred under the following circumstances:

a) **Strong Non-Linearity:** When the true relationship between the response (Y) and the predictors (X) is strongly non-linear. In such cases, non-parametric methods like KNN can substantially outperform linear regression.

b) **Prediction Accuracy is the Goal:** If prediction accuracy is the primary objective and the strong functional assumptions of the less flexible method are believed to be false.

c) **Low Dimensions:** When the number of predictors (p) is small (e.g., $p = 1$ or $p = 2$), where the curse of dimensionality is less severe.

► A **less flexible approach** might be preferred under the following circumstances:

a) **True Relationship is Simple/Parametric:** If the parametric form selected (e.g., linear model) is close to the true functional form of f, the parametric approach will tend to outperform the non-parametric approach.

b) **Interpretability is Key:** When the goal is to understand the precise influence of predictors rather than just maximizing predictive accuracy, the simple interpretation of coefficients in linear models is highly advantageous.

c) **High Dimensions or Limited Data:** When the number of predictors (p) is large, especially relative to the number of observations (n), a less flexible approach is often superior because highly flexible methods struggle due to the curse of dimensionality.

d) **Small Difference in Error:** If the test set Mean Squared Error (MSE) of the flexible method (like KNN) is only slightly lower than that of the linear model, one might prefer the simpler, more interpretable linear regression model.

6. Describe the differences between a parametric and a non-parametric statistical learning approach. What are the advantages of a parametric approach to regression or classification (as opposed to a nonparametric approach)? What are its disadvantages?

Solution: Differences Between Parametric and Non-Parametric Approaches:

Feature	**Parametric Approach** (e.g., Linear Regression)	**Non-Parametric Approach** (e.g., K-Nearest Neighbors)
Assumption of $f(X)$	Explicitly assumes a fixed functional form for $f(X)$, such as a linear relationship.	Does not explicitly assume a parametric form for f(X).
Flexibility	Less flexible, as assumptions about the functional form are strong.	More flexible, as it does not rely on predefined functional assumptions.
Parameters	Involves estimating only a small number of coefficients (parameters).	Does not estimate a fixed set of parameters characterizing the entire function.

Advantages of a Parametric Approach: The advantages of a parametric approach (like linear regression) often center on simplicity, interpretability, and performance under specific conditions:

1) **Ease of Fitting:** Parametric methods are often easy to fit, as they only require estimating a small number of coefficients.

2) **Interpretability and Inference:** Coefficients have **simple interpretations**. Furthermore, tests of statistical significance (such as computing p-values and confidence intervals) can be easily performed.

3) **Accuracy under Correct Assumptions:** The parametric approach will outperform the non-parametric approach if the parametric form that has been selected is close to the true form of f.

4) **Performance in High Dimensions/Low Data:** Parametric methods tend to outperform non-parametric approaches when there is a small number of observations per predictor. When the number of predictors (p) is large, the performance of linear regression deteriorates slowly compared to non-parametric methods like KNN, which suffer greatly from the "curse of dimensionality".

5) **Preference for Simplicity:** If the non-parametric method offers only a slightly lower test MSE (error rate), the simpler, interpretable linear model may be preferred for the sake of a model that can be described in terms of just a few coefficients.

Disadvantages of a Parametric Approach: The primary disadvantages of a parametric approach arise from its restrictive nature:

1) **Strong Assumptions (Bias):** Parametric methods inherently make strong assumptions about the form of $f(X)$.

2) **Poor Performance for Non-Linearity:** If the specified functional form (e.g., linearity) is far from the truth, and prediction accuracy is the primary goal, the parametric method will perform poorly. When the relationship is substantially non-linear, a non-parametric method like KNN regression may substantially outperform linear regression.

7. The table below provides a training data set containing six observations, three predictors, and one qualitative response variable.

Obs.	X1	X2	X3	Y
1	0	3	0	Red
2	2	0	0	Red
3	0	1	3	Red
4	0	1	2	Green
5	-1	0	1	Green
6	1	1	1	Red

Suppose we wish to use this data set to make a prediction for Y when $X1 = X2 = X3 = 0$ using K-nearest neighbors.

(a) Compute the Euclidean distance between each observation and the test point, $X1 = X2 = X3 = 0$.

(b) What is our prediction with $K = 1$? Why?

(c) What is our prediction with $K = 3$? Why?

(d) If the Bayes decision boundary in this problem is highly nonlinear, then would we expect the best value for K to be large or small? Why?

Solution: (a) Compute the Euclidean distance between each observation and the test point, $\mathbf{X} = (0, 0, 0)$:

Let observation i have predictors (x_{i1}, x_{i2}, x_{i3}). Euclidean distance:

$$d_i = \sqrt{x_{i1}^2 + x_{i2}^2 + x_{i3}^2}$$

$$\begin{aligned}
d_1 &= \sqrt{0^2 + 3^2 + 0^2} = 3 \\
d_2 &= \sqrt{2^2 + 2^2 + 0^2} = \sqrt{8} \approx 2.83 \\
d_3 &= \sqrt{0^2 + 1^2 + 3^2} = \sqrt{10} \approx 3.16 \\
d_4 &= \sqrt{0^2 + 1^2 + 2^2} = \sqrt{5} \approx 2.24 \\
d_5 &= \sqrt{(-1)^2 + 0^2 + 1^2} = \sqrt{2} \approx 1.41 \\
d_6 &= \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3} \approx 1.73
\end{aligned}$$

(b) What is our prediction with $K = 1$? Why?

The nearest neighbor is obs 5 ($d_5 = 1.41$) which has label `Green`.
Prediction: **Green**

(c) What is our prediction with $K = 3$? Why?

The 3 nearest points are: obs 5 (Green), obs 6 (Red), obs 4 (Green).
Majority is Green $(2, \text{Green}; 1, \text{Red})$.
Prediction: **Green**

(d) If the Bayes decision boundary is highly nonlinear, should K be large or small? Why?

If the Bayes boundary is highly nonlinear, a small K is preferred. A small K allows the KNN classifier to be more flexible, closely following local data structure and capturing nonlinear boundaries (low bias, high variance). A large K yields a smoother, more linear boundary (higher bias, lower variance), possibly missing nonlinear features in the data.

Chapter 3: Linear Regression

1. Describe the null hypotheses to which the p-values given in Table 3.4 correspond. Explain what conclusions you can draw based on these p-values. Your explanation should be phrased in terms of sales, TV, radio, and newspaper, rather than in terms of the coefficients of the linear model.

Solution: Table 3.4 (not provided in ISLP directly, but referenced for its content) presumably presents the results of a multiple linear regression of sales onto TV, radio, and newspaper advertising budgets, as discussed in the context of the Advertising data set. In this setting, each p-value corresponds to a null hypothesis testing whether a specific predictor has a significant effect on the response, assuming other predictors are held constant.

(a) **Null Hypothesis for TV:**

- $H_0 : \beta_{TV} = 0$
- **In terms of Sales, TV, Radio, Newspaper:** There is no relationship between the amount spent on TV advertising and product sales, assuming that the spending on radio and newspaper advertising is held constant.

(b) **Null Hypothesis for Radio:**

- $H_0 : \beta_{radio} = 0$
- **In terms of Sales, TV, Radio, Newspaper:** There is no relationship between the amount spent on radio advertising and product sales, assuming that the spending on TV and newspaper advertising is held constant.

(c) **Null Hypothesis for Newspaper:**

- $H_0 : \beta_{newspaper} = 0$
- **In terms of Sales, TV, Radio, Newspaper:** There is no relationship between the amount spent on newspaper advertising and product sales, assuming that the spending on TV and radio advertising is held constant.

Conclusions from p-values (based on typical outcomes discussed in ISLP, Section 3.4 on Marketing Plan for Advertising data):

- If the p-values for **TV** and **radio** are typically very low (e.g., < 0.05), we would reject their respective null hypotheses. This suggests that there is clear evidence of a statistically significant relationship between TV advertising budget and sales, and between radio advertising budget and sales, even when accounting for the other media.
- If the p-value for **newspaper** is typically high (e.g., > 0.05), we would not reject its null hypothesis. This implies that there is no statistically significant relationship between newspaper advertising budget and sales when TV and radio advertising budgets are already included in the model. In other words, newspaper advertising does not provide additional unique information about sales beyond what TV and radio already provide.

2. Carefully explain the differences between the KNN classifier and KNN regression methods.

Solution: K-Nearest Neighbors (KNN) is a versatile statistical learning method that can be used for both classification and regression problems, depending on the type of response variable.

(a) **KNN Classifier:**

- **Purpose:** Used to predict a **qualitative (categorical) response variable**. For example, predicting whether an email is spam (Yes/No) or classifying a tumor as malignant/benign.
- **Methodology:** For a given test observation for which we want to predict its class:
  (a) The method identifies the K training observations that are closest to the test observation in the predictor space. Distance metrics like Euclidean distance are commonly used.
  (b) It then predicts the class of the test observation by assigning it to the class that is most frequently observed among these K nearest neighbors (a majority vote).
- **Output:** A class label (e.g., "blue" or "orange" as in Figure 2.13).
- **Flexibility:** The value of K dictates the flexibility. A small K (e.g., $K = 1$) leads to a highly flexible and wiggly decision boundary, which can overfit. A large K leads to a less flexible and smoother decision boundary.

(b) **KNN Regression:**

- **Purpose:** Used to predict a **quantitative (numerical) response variable**. For example, predicting a house price or a person's income.
- **Methodology:** For a given test observation for which we want to predict its numerical response:
  (a) The method identifies the K training observations that are closest to the test observation in the predictor space, similar to the classifier.
  (b) It then predicts the response for the test observation by calculating the **average** of the response values of these K nearest neighbors.
- **Output:** A numerical value (e.g., a predicted income in thousands of dollars).
- **Flexibility:** Similar to classification, a small K results in a more flexible fit that can lead to a wiggly regression function, potentially overfitting. A large K provides a smoother, less flexible fit.

Key Differences Summarized:

- **Response Type:** KNN Classifier handles qualitative responses, while KNN Regression handles quantitative responses.
- **Prediction Rule:** The classifier uses a majority vote among neighbors, whereas the regression method uses the average of neighbors' response values.

3. Suppose we have a data set with five predictors, $X_1 = \text{GPA}$, $X_2 = \text{IQ}$, $X_3 = \text{Level}$ (1 for College and 0 for High School), $X_4 = \text{Interaction between GPA and IQ}$, and $X_5 = \text{Interaction between GPA and Level}$. The response is starting salary after graduation (in thousands of dollars). Suppose we use least squares to fit the model, and get $\hat{\beta}_0 = 50, \hat{\beta}_1 = 20, \hat{\beta}_2 = 0.07, \hat{\beta}_3 = 35, \hat{\beta}_4 = 0.01, \hat{\beta}_5 = -10$.

   (a) Predict the salary of a high school graduate with IQ of 100 and a GPA of 3.0.
   (b) Predict the salary of a college graduate with IQ of 110 and a GPA of 4.0.
   (c) True or false: Since the coefficient for the GPA/IQ interaction term is very small, there is very little evidence of an interaction effect. Justify your answer.

Solution: The linear regression model is of the form: $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \hat{\beta}_3 X_3 + \hat{\beta}_4 X_4 + \hat{\beta}_5 X_5$ Given coefficients: $\hat{\beta}_0 = 50, \hat{\beta}_1 = 20, \hat{\beta}_2 = 0.07, \hat{\beta}_3 = 35, \hat{\beta}_4 = 0.01, \hat{\beta}_5 = -10$.

(a) **Predict the salary of a high school graduate with IQ of 100 and a GPA of 3.0.**

- $X_1 = \text{GPA} = 3.0$

- $X_2 = \text{IQ} = 100$
- $X_3 = \text{Level} = 0$ (for High School)
- $X_4 = \text{GPA} \times \text{IQ} = 3.0 \times 100 = 300$
- $X_5 = \text{GPA} \times \text{Level} = 3.0 \times 0 = 0$

Substitute these values into the model: $\hat{Y} = 50 + 20(3.0) + 0.07(100) + 35(0) + 0.01(300) + (-10)(0)$ $\hat{Y} = 50 + 60 + 7 + 0 + 3 + 0$ $\hat{Y} = \mathbf{120}$ thousand dollars.

(b) **Predict the salary of a college graduate with IQ of 110 and a GPA of 4.0.**

- $X_1 = \text{GPA} = 4.0$
- $X_2 = \text{IQ} = 110$
- $X_3 = \text{Level} = 1$ (for College)
- $X_4 = \text{GPA} \times \text{IQ} = 4.0 \times 110 = 440$
- $X_5 = \text{GPA} \times \text{Level} = 4.0 \times 1 = 4.0$

Substitute these values into the model: $\hat{Y} = 50 + 20(4.0) + 0.07(110) + 35(1) + 0.01(440) + (-10)(4.0)$ $\hat{Y} = 50 + 80 + 7.7 + 35 + 4.4 - 40$ $\hat{Y} = \mathbf{137.1}$ thousand dollars.

(c) **True or false: Since the coefficient for the GPA/IQ interaction term is very small, there is very little evidence of an interaction effect. Justify your answer.**

False. The magnitude of a coefficient alone does not determine the statistical significance or the practical importance of a predictor or interaction term. The value of a coefficient depends heavily on the scale of the associated predictor. A small coefficient (like $\hat{\beta}_4 = 0.01$) could still indicate a significant effect if the predictor it's multiplied by (in this case, $X_4 = \text{GPA} \times \text{IQ}$) has a large range. For example, X_4 can be quite large (e.g., $4.0 \times 110 = 440$), making the term $0.01 \times 440 = 4.4$ a meaningful contribution to salary. To assess the evidence for an interaction effect, one should examine the **p-value** associated with the coefficient. A small p-value would indicate strong evidence against the null hypothesis that the true coefficient is zero, regardless of its magnitude.

4. I collect a set of data (n = 100 observations) containing a single predictor and a quantitative response. I then fit a linear regression model to the data, as well as a separate cubic regression, i.e. $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \epsilon$.

(a) Suppose that the true relationship between X and Y is linear, i.e. $Y = \beta_0 + \beta_1 X + \epsilon$. Consider the training residual sum of squares (RSS) for the linear regression, and also the training RSS for the cubic regression. Would we expect one to be lower than the other, would we expect them to be the same, or is there not enough information to tell? Justify your answer.

(b) Answer (a) using test rather than training RSS.

(c) Suppose that the true relationship between X and Y is not linear, but we don't know how far it is from linear. Consider the training RSS for the linear regression, and also the training RSS for the cubic regression. Would we expect one to be lower than the other, would we expect them to be the same, or is there not enough information to tell? Justify your answer.

(d) Answer (c) using test rather than training RSS.

Solution:

(a) **True relationship is linear ($Y = \beta_0 + \beta_1 X + \epsilon$). Comparing training RSS.** We would expect the **cubic regression's training RSS to be lower or equal to the linear regression's training RSS**.

Justification: A cubic regression model is more flexible than a linear regression model because it includes higher-order polynomial terms (X^2 and X^3). If the true relationship is linear, the cubic

model can simply set the coefficients for X^2 and X^3 (β_2 and β_3) to zero, effectively acting as a linear model. However, being more flexible, the cubic model also has the capacity to fit the random error (noise) in the training data more closely than the simpler linear model. Therefore, it will achieve a training RSS that is at least as good as, and typically better (lower), than the linear model.

(b) **True relationship is linear ($Y = \beta_0 + \beta_1 X + \epsilon$). Comparing test RSS.** We would expect the **linear regression's test RSS to be lower than the cubic regression's test RSS**.

Justification: Since the true relationship is linear, the linear regression model accurately reflects the underlying data-generating process (it has low bias). The cubic regression, being more flexible, will fit the noise in the training data, leading to higher variance and overfitting. This overfitting means it will perform worse on unseen test data, resulting in a higher test RSS compared to the linear model, which is simpler and less prone to capturing noise when the true relationship is simple.

(c) **True relationship is non-linear (unknown how far from linear). Comparing training RSS.** We would expect the **cubic regression's training RSS to be lower than the linear regression's training RSS**.

Justification: The cubic model is more flexible and can capture non-linear patterns that a linear model cannot. Even if the non-linearity is subtle, the cubic model will be able to approximate it better and also fit the training noise more closely, leading to a lower training RSS.

(d) **True relationship is non-linear (unknown how far from linear). Comparing test RSS. There is not enough information to tell.**

Justification: This is a trade-off scenario.

- If the true non-linear relationship is complex and well-captured by a cubic polynomial (or something close to it), the cubic regression model might have a lower test RSS than the linear model because its reduced bias outweighs its increased variance.
- However, if the true non-linear relationship is only slightly non-linear, or if the data contains a lot of noise, the increased flexibility of the cubic model might lead to overfitting the noise, causing its variance to increase significantly. In this case, the linear model, despite having higher bias, might achieve a lower test RSS due to its much lower variance.

Without knowing the specific nature of the non-linearity and the level of noise, we cannot definitively say which model would have a lower test RSS.

5. Consider the fitted values that result from performing linear regression without an intercept. In this setting, the ith fitted value takes the form $\hat{y}i = x_i\hat{\beta}$, where $\hat{\beta} = (\sum i = 1^n x_i y_i) \,/\, \left(\sum\limits_{i'=1}^{n} x_{i'}^2\right)$.
Show that we can write

$$\hat{y}i = \sum i' = 1^n a_{i'} y_{i'}.$$

What is $a_{i'}$?

Note: We interpret this result by saying that the fitted values from linear regression are linear combinations of the response values.

Solution: Given the equations for the fitted value $\hat{y}_i$ and the coefficient estimate $\hat{\beta}$ for linear regression

without an intercept:

$$(1) \quad \hat{y}_i = x_i\hat{\beta} \qquad\qquad (2) \quad \hat{\beta} = \frac{\sum_{i'=1}^{n} x_{i'}y_{i'}}{\sum_{j=1}^{n} x_j^2}$$

To show that $\hat{y}i$ can be written as a linear combination of the response values yi', we substitute the expression for $\hat{\beta}$ from (2) into (1):

$$\hat{y}_i = x_i \left(\frac{\sum_{i'=1}^{n} x_{i'}y_{i'}}{\sum_{j=1}^{n} x_j^2} \right).$$

We can rewrite the sum in the numerator by separating $y_{i'}$:

$$\hat{y}_i = x_i \left(\frac{x_1y_1 + x_2y_2 + \cdots + x_ny_n}{\sum_{j=1}^{n} x_j^2} \right).$$

This can be expressed as a sum:

$$\hat{y}_i = \frac{x_ix_1}{\sum_{j=1}^{n} x_j^2}y_1 + \frac{x_ix_2}{\sum_{j=1}^{n} x_j^2}y_2 + \cdots + \frac{x_ix_n}{\sum_{j=1}^{n} x_j^2}y_n.$$

Or, more compactly, using summation notation:

$$\hat{y}_i = \sum_{i'=1}^{n} \left(\frac{x_ix_{i'}}{\sum_{j=1}^{n} x_j^2} \right) y_{i'}$$

Comparing this to the desired form $\hat{y}_i = \sum_{i'=1}^{n} a_{i'}y_{i'}$, we can identify $a_{i'}$ as:

$$a_{i'} = \frac{x_ix_{i'}}{\sum_{j=1}^{n} x_j^2}.$$

This result demonstrates that each fitted value $\hat{y}_i$ is a weighted average of all the observed response values $y_{i'}$, where the weights $a_{i'}$ depend on both the predictor value x_i for the current observation and the predictor values $x_{i'}$ for all observations in the training set.

6. Using (3.4), argue that in the case of simple linear regression, the least squares line always passes through the point $(\bar{x}, \bar{y})$.

Solution: In simple linear regression, the model is $Y = \beta_0 + \beta_1 X + \epsilon$. The least squares estimates $\hat{\beta}_0$ and $\hat{\beta}1$ are chosen to minimize the Residual Sum of Squares (RSS), given by $\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1x_i)^2$. One of the conditions for minimizing the RSS is that the sum of the residuals must be zero. This is derived by taking the partial derivative of RSS with respect to $\hat{\beta}_0$ and setting it to zero:

$$\frac{\partial}{\partial\hat{\beta}_0} \sum_{i=1}^{n} \left(y_i - \hat{\beta}_0 - \hat{\beta}1x_i\right)^2 = 0 \implies 2\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1x_i)(-1) = 0 \implies \sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1x_i) = 0.$$

Expanding this sum:

$$\sum_{i=1}^{n} y_i - \sum_{i=1}^{n} \hat{\beta}0 - \sum_{i=1}^{n} \hat{\beta}_1 x_i = 0 \implies \sum y_i - n\hat{\beta}_0 - \hat{\beta}_1 \sum x_i = 0.$$

Now, divide the entire equation by n:

$$\frac{\sum y_i}{n} - \hat{\beta}_0 - \hat{\beta}_1 \frac{\sum x_i}{n} = 0.$$

By definition, $\bar{y} = \frac{\sum y_i}{n}$ and $\bar{x} = \frac{\sum x_i}{n}$. Substituting these into the equation: $\bar{y} - \hat{\beta}_0 - \hat{\beta}_1 \bar{x} = 0$ Rearranging this equation, we get: $\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$.

This equation is precisely the form of the fitted regression line ($\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$), but with the mean values $(\bar{x}, \bar{y})$ substituted for $(x, \hat{y})$. This demonstrates that the point $(\bar{x}, \bar{y})$ always lies on the least squares regression line, meaning the line passes through the center of the data.

7. It is claimed in the text that in the case of simple linear regression of Y onto X, the R^2 statistic

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS}, \qquad (3.17,\ ISLP)$$

is equal to the square of the correlation between X and Y

$$Cor(X, Y) = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}}. \qquad (3.18,\ ISLP)$$

Prove that this is the case. For simplicity, you may assume that $\bar{x} = \bar{y} = 0$.

Solution: The R^2 statistic is defined as:

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}} = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}.$$

Given the assumption that $\bar{x} = 0$ and $\bar{y} = 0$:

1) The Total Sum of Squares (TSS) simplifies to, $TSS = \sum_{i=1}^{n} y_i^2$ (since $\bar{y} = 0$).

2) The simple linear regression model is, $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$.

3) As shown in Exercise 6, the least squares line passes through $(\bar{x}, \bar{y})$. Since $\bar{x} = 0$ and $\bar{y} = 0$, we have $0 = \hat{\beta}_0 + \hat{\beta}_1(0)$, which implies $\hat{\beta}_0 = 0$.

4) So, the fitted model simplifies to $\hat{y}_i = \hat{\beta}_1 x_i$.

5) The least squares estimate for $\hat{\beta}_1$ (when $\bar{x} = 0, \bar{y} = 0$) is

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} x_i y_i}{\sum_{i=1}^{n} x_i^2}$$

Now, substitute these into the R^2 formula:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{\beta}_1 x_i)^2}{\sum_{i=1}^{n} y_i^2}.$$

Expand the numerator:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i^2 - 2\hat{\beta}_1 x_i y_i + \hat{\beta}_1^2 x_i^2)}{\sum_{i=1}^{n} y_i^2}$$

$$R^2 = 1 - \frac{\sum y_i^2 - 2\hat{\beta}_1 \sum x_i y_i + \hat{\beta}_1^2 \sum x_i^2}{\sum y_i^2}$$

Substitute the expression for $\hat{\beta}_1 = \frac{\sum x_i y_i}{\sum x_i^2}$:

$$\begin{aligned}
R^2 &= 1 - \frac{\sum y_i^2 - 2\left(\frac{\sum x_i y_i}{\sum x_i^2}\right)\sum x_i y_i + \left(\frac{\sum x_i y_i}{\sum x_i^2}\right)^2 \sum x_i^2}{\sum y_i^2} \\
\implies R^2 &= 1 - \frac{\sum y_i^2 - 2\frac{(\sum x_i y_i)^2}{\sum x_i^2} + \frac{(\sum x_i y_i)^2}{(\sum x_i^2)^2}\sum x_i^2}{\sum y_i^2} \\
\implies R^2 &= 1 - \frac{\sum y_i^2 - 2\frac{(\sum x_i y_i)^2}{\sum x_i^2} + \frac{(\sum x_i y_i)^2}{\sum x_i^2}}{\sum y_i^2} == 1 - \frac{\sum y_i^2 - \frac{(\sum x_i y_i)^2}{\sum x_i^2}}{\sum y_i^2} \\
&= \frac{\sum y_i^2 - \left(\sum y_i^2 - \frac{(\sum x_i y_i)^2}{\sum x_i^2}\right)}{\sum y_i^2} = \frac{\frac{(\sum x_i y_i)^2}{\sum x_i^2}}{\sum y_i^2} = \frac{(\sum x_i y_i)^2}{(\sum x_i^2)(\sum y_i^2)}
\end{aligned}$$

Now, consider the square of the correlation between X and Y. The correlation coefficient r_{XY} (or ρ_{XY}) is defined as:

$$r_{XY} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}}.$$

Given the assumption $\bar{x} = 0$ and $\bar{y} = 0$, this simplifies to:

$$r_{XY} = \frac{\sum x_i y_i}{\sqrt{(\sum x_i^2)(\sum y_i^2)}}.$$

Squaring the correlation coefficient:

$$r_{XY}^2 = \left(\frac{\sum x_i y_i}{\sqrt{(\sum x_i^2)(\sum y_i^2)}}\right)^2 = \frac{(\sum x_i y_i)^2}{(\sum x_i^2)(\sum y_i^2)}.$$

By comparing the derived expressions for R^2 and r_{XY}^2, we see that: $R^2 = r_{XY}^2$ This proves that in simple linear regression, the R^2 statistic is equal to the square of the correlation between the predictor X and the response Y, under the assumption that $\bar{x} = \bar{y} = 0$. This result holds generally even without this assumption, but the derivation is more involved.

Chapter 4: Classification

1. Using a little bit of algebra, prove that (4.2) is equivalent to (4.3). In other words, the logistic function representation and logit representation for the logistic regression model are equivalent.

Solution: The logistic function representation (4.2) for the probability $p(X) = \Pr(Y = 1|X)$ is given by:

$$p(X) = \frac{e^{\beta_0+\beta_1 X}}{1 + e^{\beta_0+\beta_1 X}}.$$

The logit representation (4.3) is given by:

$$\log\left(\frac{p(X)}{1 - p(X)}\right) = \beta_0 + \beta_1 X.$$

We will start with the logistic function representation and derive the logit representation. First, let's find an expression for $1 - p(X)$:

$$1 - p(X) = 1 - \frac{e^{\beta_0+\beta_1 X}}{1 + e^{\beta_0+\beta_1 X}} = \frac{(1 + e^{\beta_0+\beta_1 X}) - e^{\beta_0+\beta_1 X}}{1 + e^{\beta_0+\beta_1 X}} = \frac{1}{1 + e^{\beta_0+\beta_1 X}}.$$

Now, we can compute the odds ratio $\frac{p(X)}{1 - p(X)}$:

$$\frac{p(X)}{1 - p(X)} = \frac{\frac{e^{\beta_0+\beta_1 X}}{1 + e^{\beta_0+\beta_1 X}}}{\frac{1}{1 + e^{\beta_0+\beta_1 X}}} = e^{\beta_0+\beta_1 X}.$$

Finally, taking the natural logarithm of both sides:

$$\log\left(\frac{p(X)}{1 - p(X)}\right) = \log(e^{\beta_0+\beta_1 X}) = \beta_0 + \beta_1 X.$$

This algebraic manipulation shows that the logistic function representation (4.2) and the logit representation (4.3) for the logistic regression model are indeed equivalent.

2. It was stated in the text that classifying an observation to the class for which (4.17) is largest is equivalent to classifying an observation to the class for which (4.18) is largest. Prove that this is the case. In other words, under the assumption that the observations in the kth class are drawn from a $N(\mu_k, \sigma^2)$ distribution, classifying to the class for which $\pi_k f_k(x)$ is largest is equivalent to classifying to the class for which

$$\delta_k(x) = x\frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$$

is largest.

Solution: The Bayes classifier assigns an observation $X = x$ to the class k for which the posterior probability $\Pr(Y = k|X = x)$ is largest. According to Bayes' theorem:

$$\Pr(Y = k|X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^{K} \pi_l f_l(x)}.$$

Since the denominator $\sum_{l=1}^{K} \pi_l f_l(x)$ is constant for all classes k, maximizing $\Pr(Y = k|X = x)$ is equivalent to maximizing the numerator $\pi_k f_k(x)$. To simplify maximization, we can take the natural logarithm, which

is a monotonic transformation (it preserves the order of values). So, maximizing $\pi_k f_k(x)$ is equivalent to maximizing $\log(\pi_k f_k(x))$: $\log(\pi_k f_k(x)) = \log(\pi_k) + \log(f_k(x))$ Now, we use the assumption that observations in the k-th class are drawn from a normal distribution $N(\mu_k, \sigma^2)$ (univariate, with a common variance σ^2 across classes). The probability density function (PDF) for $f_k(x)$ is:

$$f_k(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu_k)^2}{2\sigma^2}\right).$$

Taking the natural logarithm of $f_k(x)$:

$$\log(f_k(x)) = \log\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) - \frac{(x-\mu_k)^2}{2\sigma^2} = -\frac{1}{2}\log(2\pi\sigma^2) - \frac{x^2 - 2x\mu_k + \mu_k^2}{2\sigma^2}.$$

Substitute this back into the expression we want to maximize:

$$\text{Maximize } \log(\pi_k) - \frac{1}{2}\log(2\pi\sigma^2) - \frac{x^2 - 2x\mu_k + \mu_k^2}{2\sigma^2}.$$

We can drop terms that are constant across all classes k without changing the result of the maximization. These constant terms are $-\frac{1}{2}\log(2\pi\sigma^2)$ and $-\frac{x^2}{2\sigma^2}$. So, maximizing is equivalent to maximizing:

$$\log(\pi_k) + \frac{2x\mu_k - \mu_k^2}{2\sigma^2} = \log(\pi_k) + \frac{x\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2}.$$

This is precisely the discriminant function $\delta_k(x)$ given in (4.18):

$$\delta_k(x) = x\frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k).$$

Therefore, classifying an observation to the class for which $\pi_k f_k(x)$ is largest is indeed equivalent to classifying it to the class for which $\delta_k(x)$ is largest.

3. A two-class problem is used to illustrate the K-nearest Neighbors classifier. The predictor space is two-dimensional. The Bayes decision boundary is known to be linear. In Section 4.5.2, Scenario 1 describes this exact situation.

   (a) If the decision boundary is highly non-linear, then would we expect the best value for K to be large or small? Why?

Solution: This question is based on conceptual understanding of KNN from ISLP, specifically regarding flexibility and K. The prompt refers to Section 4.5.2, Scenario 1, which states the Bayes decision boundary is linear. However, the conceptual question itself asks about a highly non-linear decision boundary. We will answer the question as asked, assuming a highly non-linear Bayes decision boundary.

(a) **If the decision boundary is highly non-linear, then would we expect the best value for K to be small.**
Reasons for expecting K to be small:

- A small value of K in KNN means that the classification of a test observation is determined by a very local neighborhood (just a few closest points). This results in a highly flexible decision boundary that can capture complex, non-linear patterns in the data.
- If the true Bayes decision boundary is highly non-linear, a flexible model is required to approximate it accurately (low bias). A small K allows the KNN classifier to create a very wiggly, non-linear decision boundary that can closely follow the true boundary.

- Conversely, a large value of K would average over a larger, more diverse set of neighbors, leading to a much smoother, less flexible decision boundary. This would result in high bias if the true boundary is non-linear, and thus poor performance.

It's important to note that while a small K provides flexibility and low bias for a non-linear boundary, it also increases variance, making the model sensitive to noise in the training data. The optimal K involves a balance between bias and variance, chosen via methods like cross-validation.

4. Now suppose that we wish to make a prediction for a test observation by creating a p-dimensional hypercube centered around the test observation that contains, on average, 10% of the training observations. For $p = 1, 2,$ and 100, what is the length of each side of the hypercube? Comment on your answer.

Solution: Let n be the number of training observations. We want the hypercube centered around a test observation to contain, on average, 10% of the training observations. Assume the predictor variables X_j are standardized to be in the range. In this case, the total volume of the p-dimensional predictor space is $1^p = 1$. The volume of a p-dimensional hypercube with side length L is L^p. If we want this hypercube to contain 10% of the observations, then its volume should correspond to 10% of the total volume of the predictor space. So, we set the volume of the hypercube equal to 0.10: $L^p = 0.10$ To find the side length L, we take the p-th root: $L = (0.10)^{1/p}$.

Let's calculate L for different values of p:

(a) **For $p = 1$ (1-dimensional space, a line segment):** $L = (0.10)^{1/1} = 0.10$ The length of the line segment is 0.10.

(b) **For $p = 2$ (2-dimensional space, a square):** $L = (0.10)^{1/2} = \sqrt{0.10} \approx 0.316$ The length of each side of the square is approximately 0.316.

(c) **For $p = 100$ (100-dimensional space, a hypercube):** $L = (0.10)^{1/100} \approx 0.977$ The length of each side of the hypercube is approximately 0.977.

Comment on the answer: This exercise illustrates the **curse of dimensionality**. As the number of dimensions (p) increases, the volume of space expands exponentially. To maintain a constant proportion of data (in this case, 10% of observations) within a hypercube, the side length L must drastically increase.

- In 1 dimension, $L = 0.10$ means the interval covers only 10% of the range of the predictor.
- In 2 dimensions, $L \approx 0.316$ means each side covers about 31.6% of the range, but the total area is still 10%.
- In 100 dimensions, $L \approx 0.977$ means each side must cover nearly 97.7% of the range of each predictor. This implies that the "local" neighborhood defined by the hypercube is actually spanning almost the entire range of each individual predictor.

This phenomenon indicates that in high-dimensional spaces, observations become "far apart" from each other, and the concept of "nearest neighbors" becomes less meaningful. A region meant to be "local" ends up encompassing a vast portion of the predictor space. This sparsity of data in high dimensions makes many statistical learning methods, particularly non-parametric ones like KNN, challenging because there are few truly nearby neighbors to inform a prediction.

5. We now examine the differences between LDA and QDA.

(a) If the Bayes decision boundary is linear, do we expect LDA or QDA to perform better on the training set? On the test set?

(b) If the Bayes decision boundary is non-linear, do we expect LDA or QDA to perform better on the training set? On the test set?

(c) In general, as the sample size n increases, do we expect the test prediction accuracy of QDA relative to LDA to improve, decline, or be unchanged? Why?

(d) **True or False:** Even if the Bayes decision boundary for a given problem is linear, we will probably achieve a superior test error rate using QDA rather than LDA because QDA is flexible enough to model a linear decision boundary. Justify your answer.

Solution: Background on LDA and QDA:

- **Linear Discriminant Analysis (LDA):** Assumes that the observations within each class are drawn from a multivariate Gaussian distribution with a class-specific mean vector (μ_k) but a **common covariance matrix (Σ)** across all classes. This assumption leads to a linear decision boundary.
- **Quadratic Discriminant Analysis (QDA):** Assumes that the observations within each class are drawn from a multivariate Gaussian distribution with a class-specific mean vector (μ_k) and **class-specific covariance matrices (Σ_k)**. This more flexible assumption allows for non-linear (quadratic) decision boundaries.

(a) **If the Bayes decision boundary is linear:**

- **On the training set:** We would expect **QDA** to perform slightly better or equally well. QDA is more flexible and can perfectly capture a linear boundary if it exists (by implicitly fitting similar covariance matrices for each class) and also fit the noise in the training data more closely.
- **On the test set:** We would expect **LDA** to perform better. Since the Bayes decision boundary is truly linear (meaning LDA's core assumption is met), LDA is the correct model. QDA, despite its flexibility, has more parameters to estimate (class-specific covariance matrices vs. a single common covariance matrix). With limited training data, this increased flexibility leads to higher variance in QDA, causing it to overfit the training data noise and perform worse on unseen test data compared to LDA.

(b) **If the Bayes decision boundary is non-linear:**

- **On the training set:** We would expect **QDA** to perform better. QDA is designed to capture non-linear relationships by allowing class-specific covariance matrices, making it more capable of fitting the non-linear training data patterns.
- **On the test set:** We would expect **QDA** to perform better. When the true decision boundary is non-linear, LDA's assumption of a common covariance matrix is violated, leading to high bias. QDA, with its ability to model non-linear boundaries, will have lower bias and thus better approximate the true boundary, leading to superior performance on the test set.

(c) **In general, as the sample size n increases, do we expect the test prediction accuracy of QDA relative to LDA to improve, decline, or be unchanged? Why?**

We expect the test prediction accuracy of QDA relative to LDA to **improve**.
Why: QDA is a more flexible method with more parameters to estimate than LDA. When the sample size n is small, QDA's flexibility makes it prone to high variance, leading to overfitting and poorer test performance, especially if the true boundary is linear. However, as n increases, the variance associated with QDA's estimates of the class-specific covariance matrices decreases. With enough data, QDA can accurately estimate its parameters without overfitting. If the true decision boundary is non-linear, QDA's ability to model this non-linearity (lower bias) becomes more advantageous as n grows, outweighing its increased variance. Even if the true boundary is linear, with very large n, QDA's performance will converge to LDA's, and in cases of non-linearity, it will significantly outperform LDA.

(d) **True or False: Even if the Bayes decision boundary for a given problem is linear, we will probably achieve a superior test error rate using QDA rather than LDA because QDA is flexible enough to model a linear decision boundary. Justify your answer.**

False.
Justification: While QDA is indeed flexible enough to model a linear decision boundary (it can achieve this if its class-specific covariance matrices turn out to be very similar, effectively mimicking LDA's assumption), it does so at the cost of estimating more parameters. If the true Bayes decision boundary is linear, LDA's simpler model is more appropriate. QDA's increased flexibility will lead to higher variance in its estimates, and thus it will tend to overfit the training data more than LDA. This overfitting will result in a **higher test error rate** for QDA compared to LDA, not a superior one, despite its capacity to model a linear boundary. Simpler models (like LDA) are preferred when their assumptions align with the true data-generating process, due to their lower variance.

6. Suppose we collect data for a group of students in a statistics class with variables:

$$X_1 = \text{hours studied}, \ X_2 = \text{undergrad GPA}, \ \text{and } Y = \text{receive an A}.$$

We fit a logistic regression and produce estimated coefficient, $\hat{\beta}_0 = -6, \hat{\beta}_1 = 0.05, \hat{\beta}_2 = 1$.

(a) Estimate the probability that a student who studies for 40h and has an undergrad GPA of 3.5 gets an A in the class.

(b) How many hours would the student in part (a) need to study to have a 50% chance of getting an A in the class?

Solution: The logistic regression model for the probability of receiving an A ($P(Y = \text{A}|X)$) is given by:

$$\log\left(\frac{P(Y = \text{A}|X)}{1 - P(Y = \text{A}|X)}\right) = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2.$$

Given the estimated coefficients: $\hat{\beta}_0 = -6, \hat{\beta}_1 = 0.05, \hat{\beta}_2 = 1$.

(a) **Estimate the probability that a student who studies for 40h and has an undergrad GPA of 3.5 gets an A in the class.**

Here, $X_1 = 40$ (hours studied) and $X_2 = 3.5$ (undergrad GPA). First, calculate the log-odds (logit): logit $= -6 + (0.05 \times 40) + (1 \times 3.5)$ logit $= -6 + 2 + 3.5$ logit $= -0.5$.

Now, convert the log-odds back to a probability using the logistic function:

$$\begin{aligned} P(Y = \text{A}|X) &= \frac{e^{\text{logit}}}{1 + e^{\text{logit}}} = \frac{e^{-0.5}}{1 + e^{-0.5}} \approx \frac{0.6065}{1 + 0.6065} \\ P(Y = \text{A}|X) &\approx \frac{0.6065}{1.6065} \approx \mathbf{0.3775}. \end{aligned}$$

So, the estimated probability that such a student gets an A is approximately **37.75%**.

(b) **How many hours would the student in part (a) need to study to have a 50 % chance of getting an A in the class?** A 50% chance of getting an A means $P(Y = \text{A}|X) = 0.5$. For the probability to be 0.5, the log-odds must be 0, because $\frac{e^0}{1+e^0} = \frac{1}{1+1} = 0.5$. So, we set the logit to 0: $\hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 = 0$ We want to find X_1 (hours studied), while $X_2 = 3.5$ (undergrad GPA) remains the same. $-6 + (0.05 \times X_1) + (1 \times 3.5) = 0$ $-6 + 0.05X_1 + 3.5 = 0$ $0.05X_1 - 2.5 = 0$ $0.05X_1 = 2.5$ $X_1 = \frac{2.5}{0.05}$ $X_1 = \mathbf{50}$ hours.

The student would need to study **50 hours** to have a 50% chance of getting an A in the class.

7. Suppose that we wish to predict whether a given stock will issue a dividend this year ("Yes" or "No") based on X, last year's percent profit. We examine a large number of companies and discover that the mean value of X for companies that issued a dividend was $\bar{X} = 10$, while the mean for those that didn't was $\bar{X} = 0$. In addition, the variance of X for these two sets of companies was $\hat{\sigma}^2 = 36$. Finally, 80% of companies issued dividends. Assuming that X follows a normal distribution, predict the probability that a company will issue a dividend this year given that its percentage profit was $X = 4$ last year.

Solution: This is a classification problem, and since we're assuming a normal distribution for X within each class, we can use concepts from Linear Discriminant Analysis (LDA) and Bayes' theorem. Let $Y = 1$ denote that a company issued a dividend, and $Y = 0$ denote that it did not. We are given:

- **Prior Probabilities (π_k):**
  - $\pi_1 = \Pr(Y = 1) = 0.80$ (80% of companies issued dividends)
  - $\pi_0 = \Pr(Y = 0) = 1 - 0.80 = 0.20$ (20% did not)
- **Class-specific Means (μ_k):**
  - $\mu_1 = 10$ (mean profit for companies that issued dividends)
  - $\mu_0 = 0$ (mean profit for companies that did not issue dividends)
- **Common Variance (σ^2):**
  - $\sigma^2 = 36$ (variance of profit, assumed common for both classes)
- **Observed Profit ($X = x$):**
  - $x = 4$

We need to predict $\Pr(Y = 1|X = 4)$. We use Bayes' theorem:

$$\Pr(Y = 1|X = x) = \frac{\pi_1 f_1(x)}{\pi_1 f_1(x) + \pi_0 f_0(x)}.$$

Since X is assumed to follow a normal distribution within each class, the density function $f_k(x)$ is given by:

$$f_k(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu_k)^2}{2\sigma^2}\right).$$

First, calculate $f_1(4)$ (density for companies that issued dividends with $X = 4$):

$$f_1(4) = \frac{1}{\sqrt{2\pi(36)}} \exp\left(-\frac{(4-10)^2}{2 \times 36}\right) = \frac{1}{\sqrt{72\pi}} \exp\left(-\frac{(-6)^2}{72}\right) = \frac{1}{\sqrt{72\pi}} \exp\left(-\frac{36}{72}\right)$$
$$\implies f_1(4) = \frac{1}{\sqrt{72\pi}} e^{-0.5} \approx \frac{1}{15.06} \times 0.6065 \approx 0.04027.$$

Next, calculate $f_0(4)$ (density for companies that did not issue dividends with $X = 4$):

$$f_0(4) = \frac{1}{\sqrt{2\pi(36)}} \exp\left(-\frac{(4-0)^2}{2 \times 36}\right) = \frac{1}{\sqrt{72\pi}} \exp\left(-\frac{4^2}{72}\right) = \frac{1}{\sqrt{72\pi}} \exp\left(-\frac{16}{72}\right)$$
$$\implies f_0(4) \approx \frac{1}{15.06} \times e^{-0.2222} \approx \frac{1}{15.06} \times 0.8006 \approx 0.05316.$$

Now, substitute these values into Bayes' theorem to find $\Pr(Y = 1|X = 4)$:

$$\Pr(Y = 1|X = 4) = \frac{0.80 \times 0.04027}{(0.80 \times 0.04027) + (0.20 \times 0.05316)} = \frac{0.032216}{0.032216 + 0.010632}$$
$$= \frac{0.032216}{0.042848} \approx \mathbf{0.7519}.$$

The predicted probability that a company will issue a dividend this year, given its percentage profit was $X = 4$ last year, is approximately **75.19%**.

8. Suppose that we take a data set, divide it into equally-sized training and test sets, and then try out two different classification procedures. First we use logistic regression and get an error rate of 20% on the training data and 30% on the test data. Next we use 1-nearest neighbors (i.e. K = 1) and get an average error rate (averaged over both test and training data sets) of 18%. Based on these results, which method should we prefer to use for classification of new observations? Why?

Solution: Let's analyze the given error rates for both methods:

- **Logistic Regression:**
  - Training Error Rate: 20%
  - Test Error Rate: 30%
- **1-Nearest Neighbors (K=1):**
  - Average Error Rate (Training + Test): 18%

To compare the methods fairly, we need the test error rate for 1-Nearest Neighbors. For 1-Nearest Neighbors, with $K = 1$, the method is extremely flexible. It typically makes very few, if any, errors on the training data because it simply assigns each training observation its own class. Therefore, the training error rate for 1-NN is usually very low, often close to 0%. Let TR_{1NN} be the training error rate for 1-NN and TE_{1NN} be the test error rate for 1-NN. We are given the average error rate: $\frac{TR_{1NN}+TE_{1NN}}{2} = 18\%$. If $TR_{1NN} \approx 0\%$, then: $\frac{0\%+TE_{1NN}}{2} = 18\%$ $TE_{1NN} = 2 \times 18\% = 36\%$. So, the estimated test error rate for 1-Nearest Neighbors is approximately 36%.

Comparison of Test Error Rates:

- Logistic Regression Test Error: 30%
- 1-Nearest Neighbors Test Error: $\approx 36\%$

Conclusion and Preference: Based on these results, we should prefer to use **logistic regression** for classifying new observations.

Why: The primary goal in classification (especially for prediction) is to achieve a low test error rate, as this indicates how well the model generalizes to unseen data. Logistic regression achieves a lower test error rate (30%) compared to 1-Nearest Neighbors (approximately 36%). The very high flexibility of 1-NN with $K = 1$ has led to severe overfitting on the training data (very low training error), resulting in a much higher error rate on the test data due to increased variance. Logistic regression, being a less flexible method, appears to strike a better balance between bias and variance for this particular dataset, leading to better generalization performance.

9. This problem has to do with odds.

   (a) On average, what fraction of people with an odds of 0.37 of defaulting on their credit card payment will in fact default?

   (b) Suppose that an individual has a 16% chance of defaulting on her credit card payment. What are the odds that she will default?

Solution: The odds of an event occurring are defined as the ratio of the probability of the event occurring to the probability of the event not occurring. That is, Odds $= \dfrac{P}{1-P}$, where P is the probability of the event.

(a) **On average, what fraction of people with an odds of 0.37 of defaulting on their credit card payment will in fact default?**

Given: Odds $= 0.37$
We know Odds $= \dfrac{P}{1-P}$. So, $0.37 = \dfrac{P}{1-P}$. Multiply both sides by $(1-P)$: $0.37(1-P) = P$ $0.37 - 0.37P = P$ Add $0.37P$ to both sides: $0.37 = P + 0.37P$ $0.37 = 1.37P$ $P = \dfrac{0.37}{1.37} \approx \mathbf{0.2701}$ On average, approximately **27.01%** of people with an odds of 0.37 of defaulting will in fact default.

(b) **Suppose that an individual has a 16% chance of defaulting on her credit card payment. What are the odds that she will default?**

Given: $P = 16\% = 0.16$.
Using the odds formula: Odds $= \dfrac{P}{1-P} = \dfrac{0.16}{1-0.16} = \dfrac{0.16}{0.84}$ Odds $\approx \mathbf{0.1905}$
The odds that she will default are approximately **0.1905**.

10. Equation 4.32 derived an expression for $\log\left(\dfrac{\Pr(Y=k|X=x)}{\Pr(Y=K|X=x)}\right)$ in the setting where $p > 1$, so that the mean for the kth class, μ_k, is a p-dimensional vector, and the shared covariance Σ is a $p \times p$ matrix. However, in the setting with $p = 1$, (4.32) takes a simpler form, since the means $\mu_1, \ldots, \mu_K$ and the variance σ^2 are scalars. In this simpler setting, repeat the calculation in (4.32), and provide expressions for a_k and b_{k1} in terms of $\pi_k, \pi_K, \mu_k, \mu_K$, and σ^2.

Solution: Equation (4.32) represents the log odds ratio for comparing class k to a baseline class K in a multi-class setting for generative models like LDA or QDA. For LDA, we assume a common covariance matrix Σ across classes. In the univariate case ($p = 1$), this means a common variance σ^2.

The discriminant function for LDA in the univariate case is given by (4.18):

$$\delta_k(x) = x\frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$$

We are interested in the log odds ratio $\log\left(\dfrac{\Pr(Y=k|X=x)}{\Pr(Y=K|X=x)}\right)$.

This is equivalent to $\delta_k(x) - \delta_K(x)$. Let's compute this difference:

$$\log\left(\frac{\Pr(Y=k|X=x)}{\Pr(Y=K|X=x)}\right) = \left(x\frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)\right) - \left(x\frac{\mu_K}{\sigma^2} - \frac{\mu_K^2}{2\sigma^2} + \log(\pi_K)\right).$$

Group terms involving x and terms not involving x:

$$= x\left(\frac{\mu_k}{\sigma^2} - \frac{\mu_K}{\sigma^2}\right) + (\log(\pi_k) - \log(\pi_K)) - \left(\frac{\mu_k^2}{2\sigma^2} - \frac{\mu_K^2}{2\sigma^2}\right).$$

Rearranging to match the form $a_k + b_{k1}x$:

$$= \left(\log(\pi_k) - \log(\pi_K) - \frac{\mu_k^2 - \mu_K^2}{2\sigma^2}\right) + \left(\frac{\mu_k - \mu_K}{\sigma^2}\right)x.$$

Thus, the expressions for a_k and b_{k1} are:

$$\mathbf{a_k} = \log(\pi_\mathbf{k}) - \log(\pi_\mathbf{K}) - \frac{\mu_\mathbf{k}^2 - \mu_\mathbf{K}^2}{2\sigma^2}, \qquad \mathbf{b_{k1}} = \frac{\mu_\mathbf{k} - \mu_\mathbf{K}}{\sigma^2}$$

□

11. Work out the detailed forms of a_k, b_{kj}, and b_{kjl} in (4.33). Your answer should involve $\pi_k, \pi_K, \mu_k, \mu_K, \Sigma_k$, and Σ_K.

Solution: Equation (4.33) describes the log odds ratio between class k and baseline class K for QDA, where each class has its own mean vector μ_k and covariance matrix Σ_k. The general form is:

$$\log\left(\frac{\Pr(Y=k|X=x)}{\Pr(Y=K|X=x)}\right) = a_k + \sum_{j=1}^{p} b_{kj}x_j + \sum_{j=1}^{p}\sum_{l=1}^{p} c_{kjl}x_jx_l.$$

From Bayes' theorem, maximizing $\Pr(Y=k|X=x)$ is equivalent to maximizing $\pi_k f_k(x)$. Taking the log, we aim to maximize $\log(\pi_k) + \log(f_k(x))$.

For a multivariate normal distribution, the density function $f_k(x)$ is:

$$f_k(x) = \frac{1}{(2\pi)^{p/2}|\Sigma_k|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu_k)^T\Sigma_k^{-1}(x-\mu_k)\right).$$

Taking the natural logarithm of $f_k(x)$:

$$\log(f_k(x)) = -\frac{p}{2}\log(2\pi) - \frac{1}{2}\log(|\Sigma_k|) - \frac{1}{2}(x-\mu_k)^T\Sigma_k^{-1}(x-\mu_k).$$

The log odds ratio between class k and class K is then $\log(\pi_k f_k(x)) - \log(\pi_K f_K(x))$:

$$\begin{aligned}\log\left(\frac{\Pr(Y=k|X=x)}{\Pr(Y=K|X=x)}\right) &= \left[\log(\pi_k) - \frac{1}{2}\log(|\Sigma_k|) - \frac{1}{2}(x-\mu_k)^T\Sigma_k^{-1}(x-\mu_k)\right] \\ &\quad - \left[\log(\pi_K) - \frac{1}{2}\log(|\Sigma_K|) - \frac{1}{2}(x-\mu_K)^T\Sigma_K^{-1}(x-\mu_K)\right].\end{aligned}$$

(The term $-\frac{p}{2}\log(2\pi)$ cancels out as it's common to both classes). Rearrange the terms:

$$= [\log(\pi_k) - \log(\pi_K)] - \frac{1}{2}[\log(|\Sigma_k|) - \log(|\Sigma_K|)] - \frac{1}{2}\left[(x-\mu_k)^T\Sigma_k^{-1}(x-\mu_k) - (x-\mu_K)^T\Sigma_K^{-1}(x-\mu_K)\right].$$

Now, expand the quadratic terms $(x-\mu)^T\Sigma^{-1}(x-\mu) = x^T\Sigma^{-1}x - 2\mu^T\Sigma^{-1}x + \mu^T\Sigma^{-1}\mu$:

$$= -\frac{1}{2}x^T\Sigma_k^{-1}x + \mu_k^T\Sigma_k^{-1}x - \frac{1}{2}\mu_k^T\Sigma_k^{-1}\mu_k + \frac{1}{2}x^T\Sigma_K^{-1}x - \mu_K^T\Sigma_K^{-1}x + \frac{1}{2}\mu_K^T\Sigma_K^{-1}\mu_K + \text{constant terms}$$

Let's group the terms:

(a) **Constant term (a_k):** Terms that do not depend on x:

$$a_k = \log(\pi_k) - \log(\pi_K) - \frac{1}{2}(\log(|\Sigma_k|) - \log(|\Sigma_K|)) - \frac{1}{2}(\mu_k^T\Sigma_k^{-1}\mu_k - \mu_K^T\Sigma_K^{-1}\mu_K)$$

(b) **Linear terms ($\sum_{j=1}^{p} b_{kj}x_j$):** Terms that are linear in x:

$$\mu_k^T\Sigma_k^{-1}x - \mu_K^T\Sigma_K^{-1}x = (\Sigma_k^{-1}\mu_k)^Tx - (\Sigma_K^{-1}\mu_K)^Tx = (\Sigma_k^{-1}\mu_k - \Sigma_K^{-1}\mu_K)^Tx.$$

So, b_{kj} is the j-th component of the vector $\Sigma_k^{-1}\mu_k - \Sigma_K^{-1}\mu_K$. Specifically,

$$\mathbf{b_{kj} = (\Sigma_k^{-1}\mu_k)_j - (\Sigma_K^{-1}\mu_K)_j}$$

(the j-th element of the vector difference $\Sigma_k^{-1}\mu_k - \Sigma_K^{-1}\mu_K$).

(c) **Quadratic terms** $\left(\sum_{j=1}^{p}\sum_{l=1}^{p} c_{kjl}x_jx_l\right)$**:** Terms that are quadratic in x:

$$-\frac{1}{2}x^T\Sigma_k^{-1}x + \frac{1}{2}x^T\Sigma_K^{-1}x = -\frac{1}{2}x^T(\Sigma_k^{-1} - \Sigma_K^{-1})x$$

The quadratic term $x^TAx = \sum_j \sum_l A_{jl}x_jx_l$.
If A is symmetric,

$$A_{jl}x_jx_l + A_{lj}x_lx_j = 2A_{jl}x_jx_l.$$

The coefficient c_{kjl} for x_jx_l (when $j \neq l$) is $-((\Sigma_k^{-1})_{jl} - (\Sigma_K^{-1})_{jl})$.
When $j = l$, it's $-\frac{1}{2}((\Sigma_k^{-1})_{jj} - (\Sigma_K^{-1})_{jj})$. More precisely, the quadratic term in the exponent is

$$\frac{1}{2}x^T(\Sigma_K^{-1} - \Sigma_k^{-1})x.$$

So, $\mathbf{c_{kjl} = \frac{1}{2}(\Sigma_K^{-1})_{jl} - \frac{1}{2}(\Sigma_k^{-1})_{jl}}$ (the jl-th element of $\frac{1}{2}(\Sigma_K^{-1} - \Sigma_k^{-1})$).

These expressions fully define the terms in the quadratic discriminant function, highlighting how the class-specific means and covariance matrices contribute to the non-linear decision boundary.

12. Suppose that you wish to classify an observation $X \in \mathbb{R}$ into apples and oranges. You fit a logistic regression model and find that

$$\Pr(Y = \text{orange}|X = x) = \frac{\exp(\hat{\beta}_0 + \hat{\beta}\text{orange}_0 + \hat{\alpha}\text{orange}_1 x)}{\exp(\hat{\beta}_0 + \hat{\beta}\text{orange}_0 + \hat{\alpha}\text{orange}_1 x) + \exp(\hat{\alpha}\text{apple}_0 + \hat{\alpha}\text{apple}_1 x)}.$$

(a) What is the log odds of orange versus apple in your model?

(b) What is the log odds of orange versus apple in your friend's model?

(c) Suppose that in your model, $\hat{\beta}_0 = 2$ and $\hat{\beta}\text{orange0} = 1.2, \hat{\alpha}\text{orange1} = -2, \hat{\alpha}\text{apple0} = 3, \hat{\alpha}\text{apple1} = 0.6$. What are the coefficient estimates in your model?

(d) Finally, suppose you apply both models from (d) to a data set with 2,000 test observations. What fraction of the time do you expect the predicted class labels from your model to agree with those from your friend's model? Explain your answer.

Chapter 5: Resampling Methods

1. Using basic statistical properties of the variance, as well as single variable calculus, derive (5.6). In other words, prove that α given by (5.6) does indeed minimize $Var(\alpha X + (1-\alpha)Y)$.

Solution: Let
$$Z(\alpha) = \alpha X + (1-\alpha)Y$$
denote the random return of the portfolio that invests a fraction α in asset X and a fraction $1-\alpha$ in asset Y. Using basic variance properties,
$$Var(aU + bV) = a^2 Var(U) + b^2 Var(V) + 2ab\, Cov(U, V)$$
for constants a, b and random variables U, V. Applying this to $Z(\alpha)$ with $U = X$, $V = Y$, $a = \alpha$, $b = 1-\alpha$ gives
$$Var\big(Z(\alpha)\big) = \alpha^2\sigma_X^2 + (1-\alpha)^2\sigma_Y^2 + 2\alpha(1-\alpha)\sigma_{XY},$$
where $\sigma_X^2 = Var(X)$, $\sigma_Y^2 = Var(Y)$, and $\sigma_{XY} = Cov(X, Y)$.
Expand the terms:
$$\begin{aligned} Var\big(Z(\alpha)\big) &= \alpha^2\sigma_X^2 + (1 - 2\alpha + \alpha^2)\sigma_Y^2 + 2\alpha(1-\alpha)\sigma_{XY} \\ &= \alpha^2\sigma_X^2 + \sigma_Y^2 - 2\alpha\sigma_Y^2 + \alpha^2\sigma_Y^2 + 2\alpha\sigma_{XY} - 2\alpha^2\sigma_{XY} \\ &= \alpha^2(\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}) + \alpha(-2\sigma_Y^2 + 2\sigma_{XY}) + \sigma_Y^2 \\ &= A\alpha^2 + B\alpha + C = f(\alpha), \text{ say,} \end{aligned}$$
with
$$A = \sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}, \quad B = -2\sigma_Y^2 + 2\sigma_{XY}, \quad C = \sigma_Y^2.$$
Since $A = Var(X - Y) \geq 0$, $f(\alpha)$ (being a quadratic with a positive leading coefficient) is convex and any stationary point is a global minimizer. Differentiate $f(\alpha)$:
$$f'(\alpha) = 2A\alpha + B.$$
Set $f'(\alpha) = 0$:
$$2A\alpha + B = 0 \quad \Longrightarrow \quad \alpha^* = -\frac{B}{2A} = \frac{2\sigma_Y^2 - 2\sigma_{XY}}{2(\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY})} = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}.$$
Rewrite the denominator as
$$\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY} = (\sigma_X^2 - \sigma_{XY}) + (\sigma_Y^2 - \sigma_{XY}),$$
hence
$$\alpha^* = \frac{\sigma_Y^2 - \sigma_{XY}}{(\sigma_X^2 - \sigma_{XY}) + (\sigma_Y^2 - \sigma_{XY})} = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}} = \frac{\sigma_Y^2 - \sigma_{XY}}{Var(X - Y)}.$$
This is exactly the expression labeled (5.6), and by convexity it minimizes $Var(\alpha X + (1-\alpha)Y)$.

2. We will now derive the probability that a given observation is part of a bootstrap sample. Suppose that we obtain a bootstrap sample from a set of n observations.

   (a) What is the probability that the first bootstrap observation is not the jth observation from the original sample? Justify your answer.

Solution: A bootstrap sample draws each of its n observations independently and uniformly from the original n observations, with replacement. Thus, for any draw, the probability of selecting the

jth observation is $\frac{1}{n}$, and so the probability of *not* selecting the jth observation on a single draw is

$$\Pr(\text{first bootstrap obs is not } j) = 1 - \frac{1}{n}.$$

(b) What is the probability that the second bootstrap observation is not the jth observation from the original sample?

Solution: Each position in the bootstrap sample is obtained by an independent draw from the same empirical distribution over $\{1, \ldots, n\}$. Hence the probability that the second bootstrap observation is not the jth original observation is again

$$\Pr(\text{second bootstrap obs is not } j) = 1 - \frac{1}{n}.$$

(c) Argue that the probability that the jth observation is not in the bootstrap sample is $(1 - 1/n)^n$.

Solution: The bootstrap sample consists of n independent draws (with replacement) from the n original observations. For each draw, the event "this draw is not j" has probability $1 - \frac{1}{n}$, as in parts (a) and (b). Therefore, the probability that *none* of the n draws selects the jth observation is

$$\Pr(j\text{th observation never selected}) = \left(1 - \frac{1}{n}\right)^n.$$

(d) When $n = 5$, what is the probability that the jth observation is in the bootstrap sample?

Solution: The probability that the jth observation is *not* in the bootstrap sample is $\left(1 - \frac{1}{5}\right)^5$. Thus the probability that it *is* in the bootstrap sample is

$$1 - \left(1 - \frac{1}{5}\right)^5 = 1 - \left(\frac{4}{5}\right)^5.$$

(e) When $n = 100$, what is the probability that the jth observation is in the bootstrap sample?

Solution: The same logic yields

$$\Pr(j\text{th in bootstrap}) = 1 - \left(1 - \frac{1}{100}\right)^{100}.$$

For large n, $\left(1 - \frac{1}{n}\right)^n \approx e^{-1}$, so this probability is numerically close to $1 - e^{-1} \approx 0.632$.

(f) When $n = 10,000$, what is the probability that the jth observation is in the bootstrap sample?

Solution:

$$\Pr(j\text{th in bootstrap}) = 1 - \left(1 - \frac{1}{10{,}000}\right)^{10{,}000}.$$

This is extremely close to $1 - e^{-1} \approx 0.632$ as well, since 10,000 is large and $\left(1 - \frac{1}{10{,}000}\right)^{10{,}000} \approx e^{-1}$.

3. We now review k-fold cross-validation.

(a) Explain how k-fold cross-validation is implemented.

Solution: Given a data set of size n, randomly partition the observations into k roughly equal-sized, non-overlapping subsets (folds). For $i = 1, \ldots, k$:

(1) Treat fold i as a validation set, and combine the remaining $k-1$ folds into a training set.

(2) Fit the statistical learning method using only the training set.

(3) Compute a prediction error measure (e.g. MSE for regression, misclassification rate for classification) on the held-out validation fold i; denote this error by MSE_i.

After cycling through all k folds, average the k validation errors to obtain the k-fold cross-validation estimate of the test error:

$$\text{CV}_k = \frac{1}{k}\sum_{i=1}^{k} \text{MSE}_i.$$

(b) What are the advantages and disadvantages of k-fold cross-validation relative to:

(i) The validation set approach?

(ii) LOOCV?

Solution:

(i) *Relative to the validation set approach.*

In the simple validation-set approach, the data are split once into a single training set and a single validation set; only the training subset (often about half of the data) is used to fit the model. This leads to:

- High variability in the estimated test error, because the result depends strongly on which specific split is chosen.
- Systematic overestimation of the test error, since the model is fit on fewer observations than are available in the full data set.

In contrast, k-fold CV:

- Uses multiple splits and averages the resulting errors, which reduces the variability of the test error estimate.
- Fits each model on a larger fraction of the data (approximately $(k-1)n/k$ observations), so the resulting error estimate has less bias compared to the validation-set estimate.

(ii) *Relative to LOOCV.*

Leave-one-out cross-validation (LOOCV) corresponds to $k = n$, so each training set uses $n-1$ observations, giving nearly unbiased test error estimates but potentially high variance and high computational cost. Specifically:

- LOOCV must (in general) refit the model n times, which can be computationally expensive for complex models or large n.
- Because each LOOCV training set is very similar (they differ in only one observation), the resulting fitted models are highly correlated, and the average test error estimate has relatively high variance.

k-fold CV with k small (such as 5 or 10) typically:

- Requires fitting the model only k times, so it is much more computationally efficient.
- Produces training sets that overlap less than in LOOCV, which reduces correlation among fitted models and tends to lower the variance of the estimated test error, at the cost of a modest increase in bias.

Empirically, $k = 5$ or 10 often gives a good compromise between bias and variance for the test error estimate.

4. Suppose that we use some statistical learning method to make a prediction for the response Y for a particular value of the predictor X. Carefully describe how we might estimate the standard deviation of our prediction.

Solution: Denote the predictor value of interest by x_0 and the fitted model by $\hat{f}$, so that the prediction is

$$\hat{Y}_0 = \hat{f}(x_0).$$

To estimate the *standard deviation* (or standard error) of $\hat{Y}_0$, one can use a resampling method such as the bootstrap.

A typical bootstrap-based procedure is:

(1) From the original data set of size n, draw B bootstrap samples of size n with replacement. Denote the bth bootstrap sample by $\mathcal{D}^{*(b)}$, for $b = 1, \ldots, B$.

(2) For each bootstrap sample $\mathcal{D}^{*(b)}$, refit the statistical learning method to obtain a fitted function $\hat{f}^{*(b)}$.

(3) For each $\hat{f}^{*(b)}$, compute the prediction at x_0:

$$\hat{Y}_0^{*(b)} = \hat{f}^{*(b)}(x_0), \qquad b = 1, \ldots, B.$$

(4) The empirical distribution of $\{\hat{Y}_0^{*(1)}, \ldots, \hat{Y}_0^{*(B)}\}$ approximates the sampling distribution of $\hat{Y}_0$. Hence an estimated standard deviation (standard error) of the prediction is

$$\widehat{\mathrm{SE}}(\hat{Y}_0) = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}\left(\hat{Y}_0^{*(b)} - \overline{\hat{Y}_0^{*}}\right)^2}, \quad \overline{\hat{Y}_0^{*}} = \frac{1}{B}\sum_{b=1}^{B}\hat{Y}_0^{*(b)}.$$

This procedure directly mimics the way the bootstrap is used in the chapter to estimate the variability of a parameter (such as the optimal allocation α in the investment example), but here it is applied to the predicted value at x_0. The resulting $\widehat{\mathrm{SE}}(\hat{Y}_0)$ provides an estimate of the standard deviation of the prediction due to sampling variability in the training data.

Chapter 6: Linear Model Selection and Regularization

1. We perform best subset, forward stepwise, and backward stepwise selection on a single data set. For each approach, we obtain $p+1$ models, containing $0, 1, 2, \ldots, p$ predictors. Explain your answers:

   (a) Which of the three models with k predictors has the smallest training RSS?

   Solution: For any given number of predictors k, the model identified by **best subset selection** will have the smallest training RSS.

   Explanation: Best subset selection explicitly fits all $\binom{p}{k}$ models containing exactly k predictors and then selects the one with the smallest training Residual Sum of Squares (RSS). Forward stepwise and backward stepwise selection, while also seeking to minimize RSS at each step, perform a more restricted search of the model space and are not guaranteed to find the globally best model for a given k predictors. Therefore, best subset selection, by its design, ensures the minimal training RSS for a specific k.

   (b) Which of the three models with k predictors has the smallest test RSS?

   Solution: None of these methods **guarantees** the smallest test RSS for a model with k predictors without an additional model selection step.

   Explanation: A low training RSS or high R^2 indicates a good fit to the training data, but this does not guarantee a low test error. Best subset selection, especially with a large search space, can lead to overfitting and high variance of coefficient estimates, resulting in poor performance on future data. Forward and backward stepwise methods also do not guarantee finding the best model for test error. To select the model with the smallest test RSS, one must use techniques such as a validation set approach, cross-validation, or information criteria like C_p, AIC, BIC, or adjusted R^2 to estimate test error and choose among the candidate models generated by these selection procedures.

   (c) **True or False:**

   (i) The predictors in the k-variable model identified by forward stepwise are a subset of the predictors in the $(k+1)$-variable model identified by forward stepwise selection.

   Solution: True.

   Explanation: Forward stepwise selection builds models by starting with no predictors and incrementally adding the predictor that provides the greatest additional improvement to the fit at each step. Thus, the $(k+1)$-variable model is formed by adding one predictor to the k-variable model, meaning the k-variable model is a strict subset of the $(k+1)$-variable model in terms of its predictors.

   (ii) The predictors in the k-variable model identified by backward stepwise are a subset of the predictors in the $(k+1)$-variable model identified by backward stepwise selection.

   Solution: True.

   Explanation: Backward stepwise selection starts with the full model (containing p predictors) and iteratively removes the least useful predictor one at a time. This means that the k-variable model (M_k) is obtained by removing one predictor from the $(k+1)$-variable model (M_{k+1}). Consequently, the predictors in M_k are a subset of the predictors in M_{k+1}.

   (iii) The predictors in the k-variable model identified by backward stepwise are a subset of the predictors in the $(k+1)$-variable model identified by forward stepwise selection.

   Solution: False.

   Explanation: This statement compares models from two different stepwise procedures. While each method builds nested subsets within its own process, there is no inherent guarantee that

the k predictors chosen by backward stepwise will be a subset of the $k+1$ predictors chosen by forward stepwise. The selection paths are distinct.

(iv) The predictors in the k-variable model identified by forward stepwise are a subset of the predictors in the $(k+1)$-variable model identified by backward stepwise selection.

Solution: False.
Explanation: Similar to the previous point, comparing a forward stepwise model to a backward stepwise model (even if they have different numbers of predictors) does not imply a subset relationship. Their selection processes are independent beyond the initial and final models.

(v) The predictors in the k-variable model identified by best subset are a subset of the predictors in the $(k+1)$-variable model identified by best subset selection.

Solution: False.
Explanation: Best subset selection identifies the best model for each subset size k independently. There is no constraint that the set of predictors for the best k-variable model must be a subset of the predictors for the best $(k+1)$-variable model. It is possible, as shown in an example with the Credit data set, that when moving from k to $k+1$ predictors, one of the previously selected predictors is replaced by a new one to achieve the globally best RSS for the larger size.

2. For parts (a) through (c), indicate which of i. through iv. is correct. Justify your answer.

(a) The lasso, relative to least squares, is:

i. More flexible and hence will give improved prediction accuracy when its increase in bias is less than its decrease in variance.

ii. More flexible and hence will give improved prediction accuracy when its increase in variance is less than its decrease in bias.

iii. Less flexible and hence will give improved prediction accuracy when its increase in bias is less than its decrease in variance.

iv. Less flexible and hence will give improved prediction accuracy when its increase in variance is less than its decrease in bias.

Solution: $\boxed{\text{iv}}$ Less flexible and hence will give improved prediction accuracy when its increase in bias is less than its decrease in variance.

Explanation: *The lasso applies an ℓ_1 penalty to the coefficient estimates, shrinking them towards zero and even setting some exactly to zero, which makes the model less flexible than ordinary least squares. This constraint introduces some bias but significantly reduces the variance of the coefficient estimates [82, 78, Figure 6.8]. When the reduction in variance is greater than the increase in bias, the overall test Mean Squared Error (MSE) (and thus prediction accuracy) improves, which is often the case in practice compared to a highly variable least squares fit.*

(b) Repeat (a) for ridge regression relative to least squares.

Solution: $\boxed{\text{iv}}$ Less flexible and hence will give improved prediction accuracy when its increase in bias is less than its decrease in variance.

Explanation: *Ridge regression applies an ℓ_2 penalty, shrinking coefficient estimates towards zero but typically not to exactly zero. This regularization makes ridge regression less flexible than least squares. Similar to the lasso, this reduction in flexibility introduces a small amount of bias but can lead to a substantial decrease in variance, particularly when the least squares estimates have high variance [63, 64, Figure 6.5]. When the decrease in variance outweighs the increase in bias, prediction accuracy on test observations improves.*

(c) Repeat (a) for non-linear methods relative to least squares.

Solution: $\boxed{\text{i}}$ More flexible and hence will give improved prediction accuracy when its increase in bias is less than its decrease in variance.

Explanation: *Non-linear methods, such as polynomial regression, splines, or Generalized Additive Models (GAMs), explicitly relax the linearity assumption of least squares, making them inherently more flexible. If the true relationship between predictors and response is non-linear, a linear model will have high bias. Increasing flexibility with non-linear methods can significantly reduce this bias. While increased flexibility can lead to higher variance, if the reduction in bias is greater than the increase in variance, the overall prediction accuracy improves. GAMs, for example, can capture non-linear relationships that standard linear regression would miss, potentially leading to more accurate predictions.*

3. Suppose we estimate the regression coefficients in a linear regression model by minimizing

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 \quad \text{subject to} \quad \sum_{j=1}^{p}|\beta_j| \leq s$$

for a particular value of s. For parts (a) through (e), indicate which of i. through v. is correct. Justify your answer.

(a) As we increase s from 0, the training RSS will:

i. Increase initially, and then eventually start decreasing in an inverted U shape.
ii. Decrease initially, and then eventually start increasing in a U shape.
iii. Steadily increase.
iv. Steadily decrease.
v. Remain constant.

Solution: $\boxed{\text{iv.}}$ Steadily decrease.

Explanation: *As s increases, the constraint $\sum_{j=1}^{p}|\beta_j| \leq s$ becomes less restrictive, allowing the lasso model to fit the training data more closely. A less constrained model will always achieve an equal or lower training RSS. Thus, training RSS will monotonically decrease as s increases.*

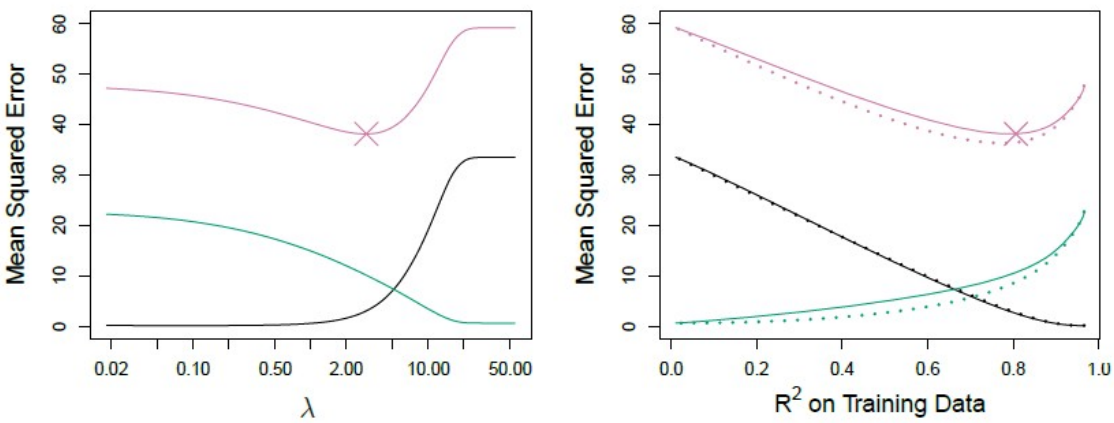


FIGURE 6.8. Left: *Plots of squared bias (black), variance (green), and test MSE (purple) for the lasso on a simulated data set.* Right: *Comparison of squared bias, variance, and test MSE between lasso (solid) and ridge (dotted). Both are plotted against their R^2 on the training data, as a common form of indexing. The crosses in both plots indicate the lasso model for which the MSE is smallest.*

(b) Repeat (a) for test RSS.

Solution: ii. Decrease initially, and then eventually start increasing in a U shape.

Explanation: *When $s = 0$, the model is the null model, typically resulting in high test RSS. As s increases, the model gains flexibility, which initially reduces bias and therefore the test RSS. However, if s becomes too large, the model becomes overly flexible and starts to overfit the training data, leading to an increase in variance and subsequently an increase in test RSS. This results in a U-shaped curve for test RSS as s increases [78, Figure 6.8].*

(c) Repeat (a) for variance.

Solution: iii. Steadily increase.

Explanation: *When $s = 0$, all β_j are zero, leading to minimal variance. As s increases, the lasso constraint loosens, allowing coefficient estimates to take on larger, more varied values (approaching the least squares estimates which can have high variance). This continuous increase in flexibility results in a steady increase in the variance of the predictions [78, Figure 6.8].*

(d) Repeat (a) for (squared) bias.

Solution: iv. Steadily decrease.

Explanation: *When $s = 0$, all β_j are zero, leading to high bias (unless all true coefficients are zero). As s increases, the lasso model becomes less constrained and its estimates move closer to the (unbiased) least squares estimates (assuming the linear model is correct). This leads to a steady decrease in bias [78, Figure 6.8].*

(e) Repeat (a) for the irreducible error.

Solution: v. Remain constant.

Explanation: *Irreducible error represents the inherent noise in the data and is independent of the model used or its tuning parameters. It cannot be reduced by improving the model fit [64, 113, 116, Figure 6.5, Figure 6.18, Figure 6.19].*

4. Suppose we estimate the regression coefficients in a linear regression model by minimizing

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}\beta_j^2$$

for a particular value of λ. For parts (a) through (e), indicate which of i. through v. is correct. Justify your answer.

(a) As we increase λ from 0, the training RSS will:

i. Increase initially, and then eventually start decreasing in an inverted U shape.
ii. Decrease initially, and then eventually start increasing in a U shape.
iii. Steadily increase.
iv. Steadily decrease.
v. Remain constant.

Solution: iii. Steadily increase.

Explanation: *When $\lambda = 0$, ridge regression is equivalent to least squares, which by definition minimizes training RSS. As λ increases, the penalty term $\lambda\sum_{j=1}^{p}\beta_j^2$ becomes more influential, forcing coefficient estimates towards zero and away from their least squares values that perfectly fit the training*

data. This increased constraint makes the model less flexible, causing the training RSS to steadily increase.

(b) Repeat (a) for test RSS.

Solution: $\boxed{\text{ii.}}$ Decrease initially, and then eventually start increasing in a U shape.

Explanation: *When $\lambda = 0$ (least squares), test RSS can be high due to high variance. As λ increases, the shrinkage reduces variance, which initially leads to a decrease in test RSS. However, if λ becomes too large, the model becomes overly biased (over-regularized), causing the test RSS to increase again [63, 64, Figure 6.5]. This results in a U-shaped curve for test RSS (or MSE) as λ increases.*

(c) Repeat (a) for variance.

Solution: $\boxed{\text{iv.}}$ Steadily decrease.

Explanation: *As λ increases, the ridge penalty explicitly shrinks the coefficient estimates towards zero. This regularization consistently reduces the flexibility of the model and the variability of the coefficient estimates, leading to a steady decrease in variance [63, 64, Figure 6.5].*

(d) Repeat (a) for (squared) bias.

Solution: $\boxed{\text{iii.}}$ Steadily increase.

Explanation: *When $\lambda = 0$ (least squares), the estimates are unbiased (assuming the linear model is correct). As λ increases, the shrinkage forces coefficients away from their true (unbiased) values, introducing and steadily increasing the bias of the estimates [63, 64, Figure 6.5].*

(e) Repeat (a) for the irreducible error.

Solution: $\boxed{\text{v.}}$ Remain constant.

Explanation: *Irreducible error is intrinsic to the data and is not affected by model choice or tuning parameters [64, 113, 116, Figure 6.5, Figure 6.18, Figure 6.19].*

5. It is well-known that ridge regression tends to give similar coefficient values to correlated variables, whereas the lasso may give quite different coefficient values to correlated variables. We will now explore this property in a very simple setting.

   Suppose $n = 2$, $p = 2$, $x_{11} = x_{12}$ and $x_{21} = x_{22}$. Furthermore, suppose that $y_1 + y_2 = 0$ and $x_{11} + x_{21} = 0$ and $x_{12} + x_{22} = 0$, so that the estimate for the intercept in a least squares, ridge regression, or lasso model is zero: $\hat{\beta}_0 = 0$.

   (a) Write out the ridge regression optimization problem in this setting.

Solution: The general ridge regression optimization problem is

$$\min_{\beta_0,\ldots,\beta_p} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^{p} \beta_j^2.$$

Given $p = 2$, $\beta_0 = 0$, and $x_{i1} = x_{i2}$, the problem simplifies to:

$$\min_{\beta_1,\beta_2} \sum_{i=1}^{n} (y_i - \beta_1 x_{i1} - \beta_2 x_{i2})^2 + \lambda(\beta_1^2 + \beta_2^2)$$

Substituting $x_{i2} = x_{i1}$:

$$\min_{\beta_1,\beta_2} \sum_{i=1}^{n} (y_i - (\beta_1 + \beta_2) x_{i1})^2 + \lambda(\beta_1^2 + \beta_2^2)$$

(b) Argue that in this setting, the ridge coefficient estimates satisfy $\hat{\beta}_1 = \hat{\beta}_2$.

Solution: In this setting, the RSS term of the objective function, $\sum_{i=1}^{n}(y_i - (\beta_1 + \beta_2)x_{i1})^2$, depends only on the sum $(\beta_1 + \beta_2)$. The penalty term, $\lambda(\beta_1^2 + \beta_2^2)$, is symmetric with respect to β_1 and β_2. Ridge regression is known to "average" or give similar coefficient values to correlated variables.

Given that x_{i1} and x_{i2} are perfectly correlated ($x_{i1} = x_{i2}$), to minimize the sum of squares of coefficients $(\beta_1^2 + \beta_2^2)$ while also achieving a desired sum $(\beta_1 + \beta_2)$ to fit the data, the optimal solution for β_1 and β_2 will be equal. If they were not equal, say $\beta_1 \neq \beta_2$, one could adjust them towards equality while keeping their sum constant, thus reducing $\beta_1^2 + \beta_2^2$ without affecting the RSS term.

Thus, ridge regression will result in $\hat{\beta}_{\mathbf{1}} = \hat{\beta}_{\mathbf{2}}$.

(c) Write out the lasso optimization problem in this setting.

Solution: The general lasso optimization problem is

$$\min_{\beta_0,\ldots,\beta_p} \sum_{i=1}^{n}(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij})^2 + \lambda \sum_{j=1}^{p}|\beta_j|.$$

Given $p = 2$, $\beta_0 = 0$, and $x_{i1} = x_{i2}$, the problem simplifies to:

$$\min_{\beta_1,\beta_2} \sum_{i=1}^{n}(y_i - \beta_1 x_{i1} - \beta_2 x_{i2})^2 + \lambda(|\beta_1| + |\beta_2|)$$

Substituting $x_{i2} = x_{i1}$:

$$\min_{\beta_1,\beta_2} \sum_{i=1}^{n}(y_i - (\beta_1 + \beta_2)x_{i1})^2 + \lambda(|\beta_1| + |\beta_2|)$$

(d) Argue that in this setting, the lasso coefficients $\hat{\beta}_1$ and $\hat{\beta}_2$ are not unique—in other words, there are many possible solutions to the optimization problem in (c). Describe these solutions.

Solution: Similar to ridge regression, the RSS term, $\sum_{i=1}^{n}(y_i - (\beta_1 + \beta_2)x_{i1})^2$, depends only on the sum $(\beta_1 + \beta_2)$. Let this optimal sum be $S^* = \hat{\beta}_1 + \hat{\beta}_2$. The lasso penalty is $|\beta_1| + |\beta_2|$.

The objective is to minimize RSS, which implies minimizing the sum S^*. Then, among all pairs (β_1, β_2) such that $\beta_1 + \beta_2 = S^*$, we need to find the one that minimizes $|\beta_1| + |\beta_2|$.

The lasso, unlike ridge regression, performs variable selection by forcing some coefficients to be exactly zero, particularly in the presence of correlated predictors. The ℓ_1 penalty (diamond-shaped constraint region in 2D) has "corners" on the axes, encouraging solutions where coefficients are zero (see Figure 6.7, ITSL).

If $S^* \neq 0$, then there are multiple solutions that minimize $|\beta_1| + |\beta_2|$ given $\beta_1 + \beta_2 = S^*$. For example, if $S^* = K$, then $(\beta_1 = K, \beta_2 = 0)$ and $(\beta_1 = 0, \beta_2 = K)$ both result in $|\beta_1| + |\beta_2| = |K|$. Any combination such that $\beta_1 + \beta_2 = K$ and one of them is zero will result in the same minimal ℓ_1 penalty for that sum. The lasso will tend to pick one of the correlated variables and set the other to zero, but which one is chosen is arbitrary when they are perfectly correlated and symmetric in their effect on RSS.

Thus, the specific choice of which variable gets the non-zero coefficient is not unique.

6. (a) Consider (6.12) with $p = 1$. For some choice of y_1 and $\lambda > 0$, plot (6.12) as a function of β_1. Your plot should confirm that (6.12) is solved by (6.14).

Solution: For $p = 1$, equation (6.12) is the ridge objective, whose minimizer is given by (6.14). (See the shrinkage section in Chapter 6 for the one-parameter ridge solution.)
Take, for concreteness, $y_1 = 1$ and $\lambda = 1$. Then (6.12) becomes

$$f(\beta_1) = (y_1 - \beta_1)^2 + \lambda\beta_1^2 = (1 - \beta_1)^2 + \beta_1^2.$$

This is a smooth convex quadratic function of β_1.

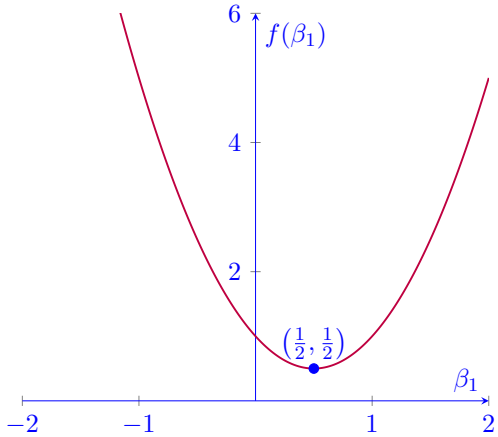
Differentiate to find the minimizer:

$$\frac{d}{d\beta_1} f(\beta_1) = -2(y_1 - \beta_1) + 2\lambda\beta_1 = -2y_1 + 2(1 + \lambda)\beta_1.$$

Setting this derivative equal to zero yields

$$\hat{\beta}_1 = \frac{y_1}{1 + \lambda},$$

which coincides with (6.14) for $j = 1$.



If $f(\beta_1)$ is plotted against β_1, the curve attains its minimum at this value $\hat{\beta}_1$, visually confirming the analytic solution.

(b) Consider (6.13) with $p = 1$. For some choice of y_1 and $\lambda > 0$, plot (6.13) as a function of β_1. Your plot should confirm that (6.13) is solved by (6.15).

Solution: For $p = 1$, equation (6.13) is the lasso objective, whose minimizer is given in (6.15) by the soft-thresholding rule. (See the lasso subsection where the coordinate-wise soft-threshold solution is derived.) Again choose $y_1 = 1$ and $\lambda = 1$. Then (6.13) becomes

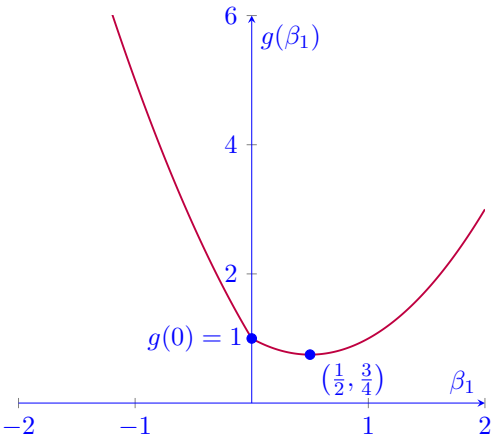
$$g(\beta_1) = (y_1 - \beta_1)^2 + \lambda|\beta_1| = (1 - \beta_1)^2 + |\beta_1|.$$

This objective is convex but has a kink at $\beta_1 = 0$ because of the absolute-value term.
The one-dimensional lasso solution (6.15) can be written as

$$\hat{\beta}_1 = \begin{cases} y_1 - \dfrac{\lambda}{2}, & y_1 > \dfrac{\lambda}{2}, \\ 0, & |y_1| \le \dfrac{\lambda}{2}, \\ y_1 + \dfrac{\lambda}{2}, & y_1 < -\dfrac{\lambda}{2}, \end{cases}$$

which is the familiar soft-thresholding operator. For $y_1 = 1$ and $\lambda = 1$, this gives

$$\hat{\beta}_1 = 1 - \frac{1}{2} = \frac{1}{2}.$$



If $g(\beta_1)$ is plotted against β_1, the resulting curve has a unique minimum at $\beta_1 = 1/2$, in agreement with the soft-threshold solution, thereby confirming that (6.13) is solved by (6.15).

7. We will now derive the Bayesian connection to the lasso and ridge regression discussed in Section 6.2.2.

(a) Suppose that $y_i = \beta_0 + \sum_{j=1}^{p} \beta_j x_{ij} + \epsilon_i$, where ϵ_i are independent and identically distributed from a $N(0, \sigma^2)$ distribution. Write out the likelihood for the data.

Solution: The assumption $\epsilon_i \sim N(0, \sigma^2)$ implies that y_i follows a normal distribution:

$$y_i \sim N\left(\beta_0 + \sum_{j=1}^{p} \beta_j x_{ij}, \sigma^2\right)$$

The likelihood for a single observation y_i is

$$f(y_i|X_i, \beta, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - (\beta_0 + \sum_{j=1}^{p} \beta_j x_{ij}))^2}{2\sigma^2}\right).$$

For n independent observations, the likelihood $f(Y|X, \beta, \sigma^2)$ is the product of individual likelihoods:

$$\mathbf{f(Y|X, \beta, \sigma^2) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij})^2\right)}$$

(b) Assume the following prior for $\beta_1, \ldots, \beta_p$: are independent and identically distributed according to a double-exponential distribution with mean 0 and common scale parameter b, i.e. $p(\beta_j) = \frac{1}{2b}\exp(-|\beta_j|/b)$. Write out the posterior for β in this setting.

Solution: From Bayes' theorem, the posterior distribution is proportional to the likelihood multiplied by the prior: $p(\beta|X, Y) \propto f(Y|X, \beta)p(\beta)$. Assuming the prior for β_0 is flat and constant, and only considering $\beta = (\beta_1, \ldots, \beta_p)^T$, the prior for β is

$$p(\beta) = \prod_{j=1}^{p} p(\beta_j) = \left(\frac{1}{2b}\right)^p \exp\left(-\frac{1}{b}\sum_{j=1}^{p}|\beta_j|\right).$$

Combining the likelihood (ignoring constant terms related to σ^2 and 2π) with this prior:

$$\begin{aligned} p(\beta|X, Y) &\propto \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2\right)\exp\left(-\frac{1}{b}\sum_{j=1}^{p}|\beta_j|\right) \\ &= \exp\left(-\frac{1}{2\sigma^2}\left[\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \frac{2\sigma^2}{b}\sum_{j=1}^{p}|\beta_j|\right]\right) \end{aligned}$$

(c) Argue that the lasso estimate is the mode for β under this posterior distribution.

Solution: The posterior mode for β is the value of β that maximizes $p(\beta|X, Y)$. Maximizing the posterior is equivalent to maximizing its logarithm. Taking the logarithm of the posterior distribution derived in (b), and discarding constant terms:

$$\log p(\beta|X, Y) \propto -\left[\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \frac{2\sigma^2}{b}\sum_{j=1}^{p}|\beta_j|\right]$$

Maximizing this is equivalent to minimizing the term in the square brackets. If we let $\lambda = \frac{2\sigma^2}{b}$, the

minimization problem becomes:

$$\min_{\beta_0,\ldots,\beta_p} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^{p} |\beta_j|$$

This is precisely the definition of the lasso optimization problem. Therefore, the **lasso estimate is the posterior mode for** β when a double-exponential (Laplace) prior is assumed for the coefficients.

(d) Now assume the following prior for $\beta_1, \ldots, \beta_p$: are independent and identically distributed according to a normal distribution with mean zero and variance c. Write out the posterior for β in this setting.

Solution: The prior for each β_j is $p(\beta_j) = \frac{1}{\sqrt{2\pi c}} \exp\left(-\frac{\beta_j^2}{2c}\right)$. The combined prior for β is

$$p(\beta) = \prod_{j=1}^{p} p(\beta_j) = \left(\frac{1}{\sqrt{2\pi c}}\right)^p \exp\left(-\frac{1}{2c}\sum_{j=1}^{p}\beta_j^2\right).$$

Combining the likelihood from (a) with this Gaussian prior (ignoring constant terms):

$$\mathbf{p(\beta|X,Y) \propto \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij})^2\right) \exp\left(-\frac{1}{2c}\sum_{j=1}^{p}\beta_j^2\right)}$$

$$\mathbf{= \exp\left(-\frac{1}{2\sigma^2}\left[\sum_{i=1}^{n}(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij})^2 + \frac{\sigma^2}{c}\sum_{j=1}^{p}\beta_j^2\right]\right)}$$

(e) Argue that the ridge regression estimate is both the mode and the mean for β under this posterior distribution.

Solution: Similar to (c), the posterior mode is found by maximizing $\log p(\beta|X,Y)$. Taking the logarithm of the posterior distribution from (d) and discarding constant terms:

$$\log p(\beta|X,Y) \propto -\left[\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \frac{\sigma^2}{c}\sum_{j=1}^{p}\beta_j^2\right]$$

Maximizing this is equivalent to minimizing the term in the square brackets. If we let $\lambda = \frac{\sigma^2}{c}$, the minimization problem becomes:

$$\min_{\beta_0,\ldots,\beta_p} \sum_{i=1}^{n}(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij})^2 + \lambda\sum_{j=1}^{p}\beta_j^2$$

This is precisely the definition of the ridge regression optimization problem. Therefore, the **ridge regression estimate is the posterior mode for** β when a Gaussian prior is assumed for the coefficients. Furthermore, when both the likelihood (errors are Gaussian) and the prior distribution (for β_j) are Gaussian, the resulting posterior distribution for β is also Gaussian. For any Gaussian distribution, its mean, median, and mode are all identical. Thus, the ridge regression estimate is also the **posterior mean** for β in this setting.

Chapter 7: Moving Beyond Linearity

1. It was mentioned in this chapter that a cubic regression spline with one knot at ξ can be obtained using a basis of the form $x, x^2, x^3, (x-\xi)_+^3$, where $(x-\xi)_+^3 = (x-\xi)^3$ if $x > \xi$ and equals 0 otherwise. We will now show that a function of the form

$$f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 (x-\xi)_+^3$$

is indeed a cubic regression spline, regardless of the values of $\beta_0, \beta_1, \beta_2, \beta_3, \beta_4$.

(a) Find a cubic polynomial

$$f_1(x) = a_1 + b_1 x + c_1 x^2 + d_1 x^3$$

such that $f(x) = f_1(x)$ for all $x < \xi$. Express a_1, b_1, c_1, d_1 in terms of the β's.

Solution: For $x < \xi$, $(x-\xi)_+^3 = 0$, so

$$f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3.$$

Hence

$$a_1 = \beta_0, \quad b_1 = \beta_1, \quad c_1 = \beta_2, \quad d_1 = \beta_3.$$

(b) Find a cubic polynomial

$$f_2(x) = a_2 + b_2 x + c_2 x^2 + d_2 x^3$$

such that $f(x) = f_2(x)$ for all $x > \xi$. Express a_2, b_2, c_2, d_2 in terms of the β's.

Solution: For $x > \xi$, $(x-\xi)_+^3 = (x-\xi)^3 = x^3 - 3\xi x^2 + 3\xi^2 x - \xi^3$, so

$$f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 (x^3 - 3\xi x^2 + 3\xi^2 x - \xi^3).$$

Collecting powers of x,

$$a_2 = \beta_0 - \beta_4 \xi^3, \quad b_2 = \beta_1 + 3\beta_4 \xi^2, \quad c_2 = \beta_2 - 3\beta_4 \xi, \quad d_2 = \beta_3 + \beta_4.$$

(c) Show that $f_1(\xi) = f_2(\xi)$; that is, $f(x)$ is continuous at ξ.

Solution: Using the expressions above,

$$f_1(\xi) = \beta_0 + \beta_1 \xi + \beta_2 \xi^2 + \beta_3 \xi^3,$$
$$f_2(\xi) = a_2 + b_2 \xi + c_2 \xi^2 + d_2 \xi^3 = \big(\beta_0 - \beta_4 \xi^3\big) + \big(\beta_1 + 3\beta_4 \xi^2\big)\xi + \big(\beta_2 - 3\beta_4 \xi\big)\xi^2 + \big(\beta_3 + \beta_4\big)\xi^3,$$

and the β_4 terms cancel, giving $f_2(\xi) = f_1(\xi)$. So f is continuous at ξ.

(d) Show that $f_1'(\xi) = f_2'(\xi)$; that is, $f'(x)$ is continuous at ξ.

Solution: For $x < \xi$, $f_1'(x) = b_1 + 2c_1 x + 3d_1 x^2 = \beta_1 + 2\beta_2 x + 3\beta_3 x^2$. Thus

$$f_1'(\xi) = \beta_1 + 2\beta_2 \xi + 3\beta_3 \xi^2.$$

For $x > \xi$, $f_2'(x) = b_2 + 2c_2 x + 3d_2 x^2$. Plugging in b_2, c_2, d_2 and $x = \xi$, the β_4 terms cancel and

$$f_2'(\xi) = \beta_1 + 2\beta_2 \xi + 3\beta_3 \xi^2 = f_1'(\xi).$$

So $f'(x)$ is continuous at ξ.

(e) Show that $f_1''(\xi) = f_2''(\xi)$; that is, $f''(x)$ is continuous at ξ. Therefore, $f(x)$ is a cubic spline.

Solution: For $x < \xi$, $f_1''(x) = 2c_1 + 6d_1 x = 2\beta_2 + 6\beta_3 x$, so

$$f_1''(\xi) = 2\beta_2 + 6\beta_3 \xi.$$

For $x > \xi$, $f_2''(x) = 2c_2 + 6d_2 x$, giving

$$f_2''(\xi) = 2(\beta_2 - 3\beta_4 \xi) + 6(\beta_3 + \beta_4)\xi = 2\beta_2 + 6\beta_3 \xi = f_1''(\xi).$$

So $f''(x)$ is continuous at ξ, and f is a cubic spline made of two cubic pieces joined smoothly at the knot.

2. A curve $\hat{g}$ is computed as

$$\hat{g} = \arg\min_g \left\{ \sum_{i=1}^{n} (y_i - g(x_i))^2 + \lambda \int [g^{(m)}(x)]^2 \, dx \right\},$$

where $g^{(m)}$ is the mth derivative ($g^{(0)} = g$). Sketch $\hat{g}$ in each case.

(a) $\lambda = \infty$, $m = 0$.

Solution: The penalty is on $\int g(x)^2 dx$, so with $\lambda = \infty$ the only allowed function is $g(x) \equiv 0$, a flat line at zero, ignoring the data completely.

(b) $\lambda = \infty$, $m = 1$.

Solution: The penalty is on $\int [g'(x)]^2 dx$, so $\lambda = \infty$ forces $g'(x) \equiv 0$, i.e. g must be constant; the constant is chosen to be roughly the average of the y_i, so the sketch is a horizontal line fit.

(c) $\lambda = \infty$, $m = 2$.

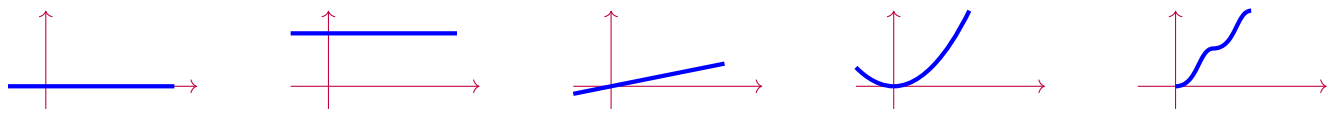
Solution: Now the penalty is on curvature; $\lambda = \infty$ forces $g''(x) \equiv 0$, so g must be a straight line (degree-1 polynomial) chosen to best fit the data, i.e. the usual simple linear regression line.

(d) $\lambda = \infty$, $m = 3$.

Solution: Here the third derivative is penalized, so $\lambda = \infty$ forces $g^{(3)}(x) \equiv 0$; g must be a quadratic (degree-2 polynomial) fitted to the points, i.e. a smooth parabola through the data.

(e) $\lambda = 0$, $m = 3$.

Solution: With $\lambda = 0$ there is no smoothness penalty, so g is chosen only to make the residual sum of squares zero; it becomes a very wiggly interpolating curve that passes exactly through all data points.



(a) $\lambda = \infty$, $m = 0$ (b) $\lambda = \infty$, $m = 0$ (c) $\lambda = \infty$, $m = 1$ (d) $\lambda = \infty$, $m = 2$ (e) $\lambda = 0$, $m = 3$

3. Basis functions $b_1(X) = X$, $b_2(X) = (X-1)^2 I(X \geq 1)$, and

$$Y = \beta_0 + \beta_1 b_1(X) + \beta_2 b_2(X) + \varepsilon,$$

with estimates $\hat{\beta}_0 = 1$, $\hat{\beta}_1 = 1$, $\hat{\beta}_2 = -2$. Sketch the fitted curve on $X \in [-2, 2]$.

Solution: For $X < 1$, $b_2(X) = 0$, so

$$\hat{f}(X) = 1 + 1 \cdot X = 1 + X, \quad X < 1,$$

a straight line with slope 1 and intercept 1 crossing zero at $X = -1$. For $X \geq 1$,

$$\hat{f}(X) = 1 + X - 2(X-1)^2,$$

which is a downward-opening quadratic joined continuously to the line at $X = 1$; at $X = 1$ the value is 2, and for larger X the curve bends down.

4. Basis functions

$$b_1(X) = I(0 \leq X \leq 2) - (X-1)I(1 \leq X \leq 2),$$
$$b_2(X) = (X-3)I(3 \leq X \leq 4) + I(4 < X \leq 5),$$

and

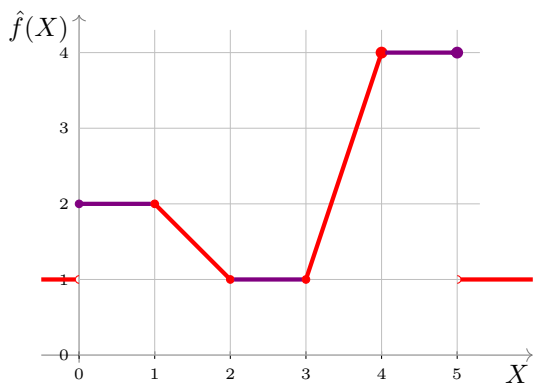
$$Y = \beta_0 + \beta_1 b_1(X) + \beta_2 b_2(X) + \varepsilon,$$

with $\hat{\beta}_0 = 1$, $\hat{\beta}_1 = 1$, $\hat{\beta}_2 = 3$. Sketch $\hat{f}(X)$ on $[-2, 6]$.

Solution: The function $\hat{f}(X)$ can be written as

$$\hat{f}(X) = \begin{cases} 1, & X < 0, \\ 2, & 0 \leq X < 1, \\ 3 - X, & 1 \leq X \leq 2, \\ 1, & 2 < X < 3, \\ 3X - 8, & 3 \leq X \leq 4, \\ 4, & 4 < X \leq 5, \\ 1, & X > 5. \end{cases}$$

Region	X-range	b_1	b_2	$1 + b_1 + 3b_2$	$\hat{f}(X)$
1	$X < 0$	0	0	$1 + 0 + 3 * 0$	1
2	$0 \leq X < 1$	1	0	$1 + 1 + 3 * 0$	2
3	$1 \leq X \leq 2$	$2 - X$	0	$1 + (2 - X) + 3 * 0$	$3 - X$
4	$2 < X < 3$	0	0	$1 + 0 + 3 * 0$	1
5	$3 \leq X \leq 4$	0	$X - 3$	$1 + 0 + 3 * (X - 3)$	$3X - 8$
6	$4 < X \leq 5$	0	1	$1 + 0 + 3 * 1$	4
7	$X > 5$	0	0	$1 + 0 + 3 * 0$	1



Overall, the curve is piecewise linear/constant, with visible "steps" and "ramps" over $[0, 2]$ and $[3, 5]$.

5. Two curves

$$\hat{g}_1 = \arg\min_g \left\{ \sum_{i=1}^{n} (y_i - g(x_i))^2 + \lambda \int [g^{(3)}(x)]^2 \, dx \right\},$$

$$\hat{g}_2 = \arg\min_g \left\{ \sum_{i=1}^{n} (y_i - g(x_i))^2 + \lambda \int [g^{(4)}(x)]^2 \, dx \right\}.$$

(a) As $\lambda \to \infty$, which has smaller training RSS?

Solution: As $\lambda \to \infty$, both fits are forced to be very smooth polynomials of low degree (third–derivative penalty gives at most quadratic; fourth–derivative penalty gives at most cubic), and each is chosen to minimize RSS within its class; the cubic class is more flexible than the quadratic class, so $\hat{g}_2$ can match the data at least as well and thus has training RSS no larger (typically smaller).

(b) As $\lambda \to \infty$, which has smaller test RSS?

Solution: With extremely large λ, both curves are very simple low-degree polynomials with high bias but low variance; the more restricted quadratic model $\hat{g}_1$ is less flexible and usually gives lower variance than the cubic, so $\hat{g}_1$ will typically have smaller test RSS.

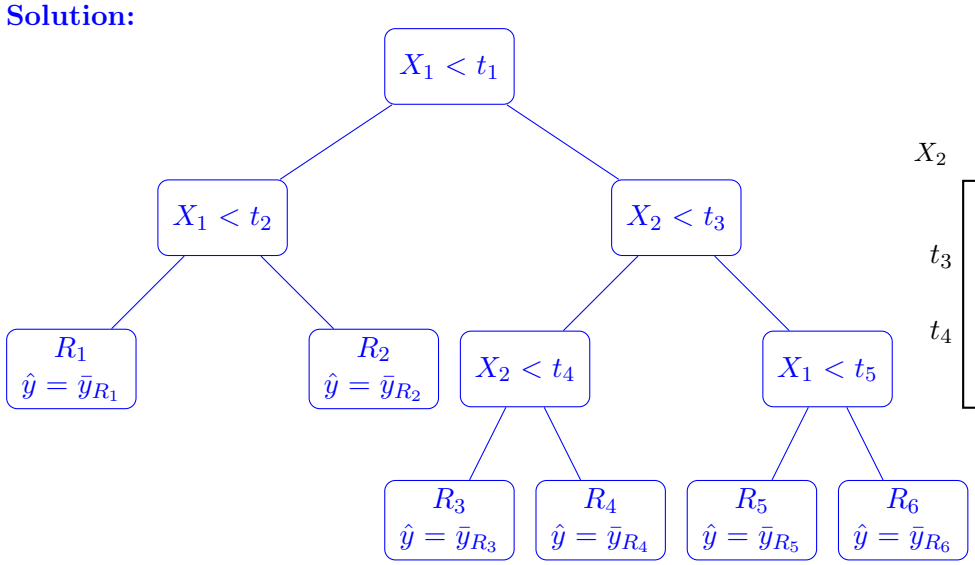
(c) For $\lambda = 0$, which has smaller training and test RSS?

Solution: With $\lambda = 0$ there is no penalty, so both problems reduce to finding the interpolating function that minimizes the training RSS, which can be driven essentially to zero in each case; the training RSS is therefore (almost) the same, but $\hat{g}_1$ and $\hat{g}_2$ will be very wiggly and overfit, so test RSS will be large and very similar, with no systematic advantage for either.

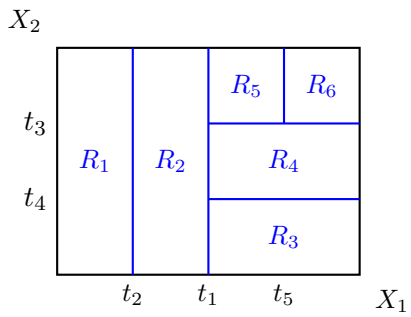
Chapter 8: Tree Based Methods

1. Draw an example (of your own invention) of a partition of two-dimensional feature space that could result from recursive binary splitting. Your example should contain at least six regions. Draw a decision tree corresponding to this partition. Be sure to label all aspects of your figures, including the regions $R_1, R_2, \ldots$, the cutpoints $t_1, t_2, \ldots$, and so forth.

Solution:



A decision tree diagram



Partition diagram

2. It is mentioned in Section **8.2.3** that boosting using depth-one trees (or stumps) leads to an additive model: that is, a model of the form

$$f(x) = \sum_{j=1}^{P} f_j(X_j),$$

Explain why this is the case. You can begin with (**8.12**) in Algorithm **8.2**.

Solution: Boosting with decision stumps (depth–one trees) gives an additive model because each stump depends on only *one* predictor, and the boosted model is a sum of many stumps.

In Algorithm 8.2, the boosted regression function after B iterations is

$$\hat{f}(x) = \sum_{b=1}^{B} \lambda \, \hat{f}_b(x),$$

where each $\hat{f}_b$ is a small tree fit to the current residuals, and λ is the shrinkage parameter (step size).

If each tree is a stump (one split, two leaves), then $\hat{f}_b(x)$ is a function of a single variable only, say $X_{j(b)}$, because the stump uses just one predictor for its single split. In other words,

$$f_b(x) = g_b\big(X_{j(b)}\big)$$

for some simple piecewise–constant function g_b. Summing many such terms gives

$$f(x) = \sum_{b=1}^{B} g_b\big(X_{j(b)}\big),$$

which is an *additive model*: it is a sum of functions, each of which depends on one coordinate of x only. This is why boosting with depth–one trees produces an additive model.

3. Consider the Gini index, classification error, and entropy in a simple classification setting with two classes. Create a single plot that displays each of these quantities as a function of $\hat{p}_{m1}$. The x-axis should display $\hat{p}_{m1}$, ranging from 0 to 1, and the y-axis should display the value of the Gini index, classification error, and entropy.

Solution: We compare three impurity measures for a binary classification problem (two classes), as a function of $p = p_{m1}$, the probability of class 1 in node m. Therefore, Class 2 has probability $p_{m2} = 1 - p$.

The three quantities are - Classification error, Gini index, and Entropy.

1) **Classification error:**

$$E(p) = 1 - \max(p,\ 1 - p).$$

This is $1 - p$ if $p \geq 0.5$ (class 1 majority) and p if $p < 0.5$ (class 2 majority). It is 0 at $p = 0$ and $p = 1$, and reaches its maximum 0.5 at $p = 0.5$.

2) **Gini index:**

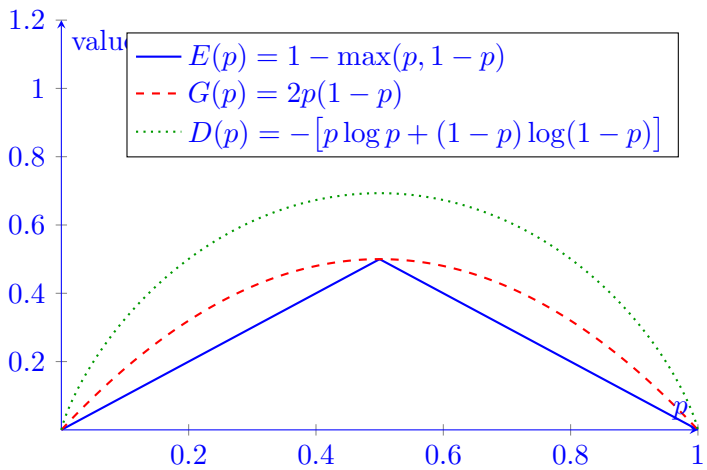
$$G(p) = p(1 - p) + (1 - p)p = 2p(1 - p).$$

This is 0 at $p = 0$ and $p = 1$, and has its maximum 0.5 at $p = 0.5$.

3) **Entropy:**

$$D(p) = -\big[p \log p + (1 - p) \log(1 - p)\big],$$

with the convention $0 \log 0 = 0$. Entropy is also 0 at $p = 0$ and $p = 1$, and reaches its maximum at $p = 0.5$.



If we plot all three against $p \in [0, 1]$:

- All are symmetric around $p = 0.5$, and all are 0 at 0 and 1.
- Classification error is "flat–topped" and less sensitive; Gini and entropy are smooth, more curved, and more sensitive to changes near $p = 0.5$.

This is why Gini and entropy are usually preferred for growing trees, while classification error is often used for pruning and final evaluation.

4. Consider the following figure:
   (a) Sketch the tree corresponding to the partition of the predictor space illustrated in the left-hand panel of Figure 2. The numbers inside the boxes indicate the mean of Y within each region.
   (b) Create a diagram similar to the left-hand panel of Figure 2, using the tree illustrated in the right-hand panel of the same figure. You should divide up the predictor space into the correct regions, and indicate the mean for each region.

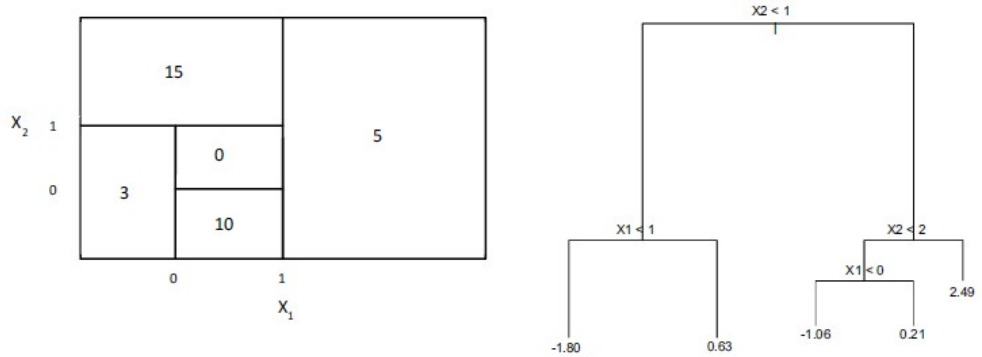
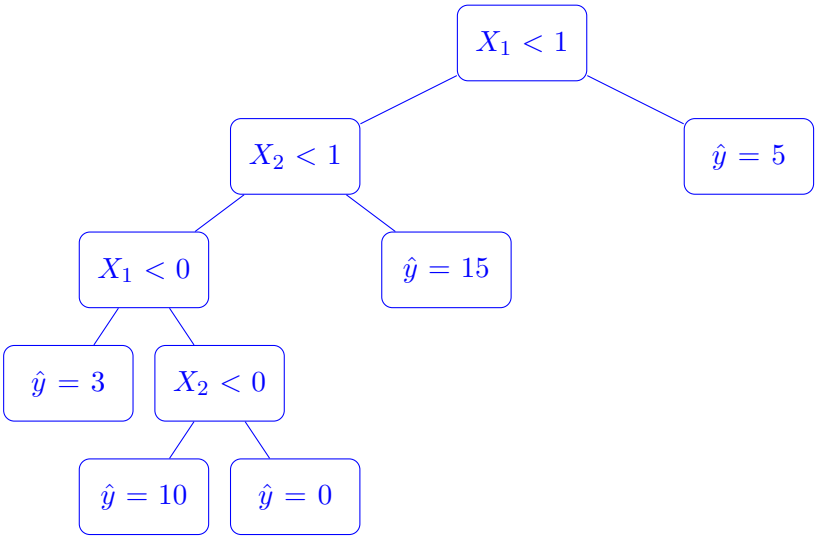


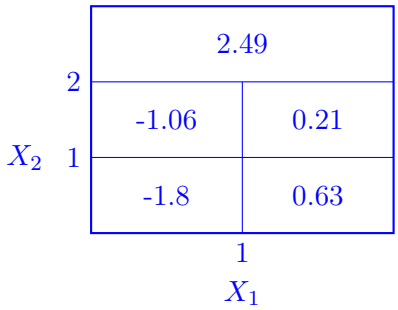
Figure 2: **Left:** A partition of the predictor space corresponding to Exercise 4a. **Right:** A tree corresponding to Exercise 4b.

Solution: This question refers to the given Figure (two plots: a partition of the predictor space and the corresponding tree). The key ideas are:

(a) **Tree from partition.**



(b) **Partition from tree.**



5. Suppose we produce ten bootstrapped samples from a data set containing red and green classes. We then apply a classification tree to each bootstrapped sample and, for a specific value of X, produce 10 estimates of $P(\text{Class is Red}|X)$:

$$0.1, 0.15, 0.2, 0.2, 0.55, 0.6, 0.6, 0.65, 0.7, \text{ and } 0.75.$$

There are two common ways to combine these results together into a single class prediction. One is the majority vote approach discussed in this chapter. The second approach is to classify based on the average probability. In this example, what is the final classification under each of these two approaches?

Solution: We have ten estimated probabilities that the class is Red at a fixed X:

$$0.10,\ 0.15,\ 0.20,\ 0.20,\ 0.55,\ 0.60,\ 0.60,\ 0.65,\ 0.70,\ 0.75.$$

Majority vote. Each tree makes a hard class prediction: Red if $P(\text{Red} \mid X) \geq 0.5$, otherwise Green .

- Values below 0.5: $0.10, 0.15, 0.20, 0.20$ (4 votes for Green).
- Values at or above 0.5: $0.55, 0.60, 0.60, 0.65, 0.70, 0.75$ (6 votes for Red).

So the majority vote classification is **Red**.

Average probability. Compute the mean:

$$\bar{p} = \frac{0.10 + 0.15 + 0.20 + 0.20 + 0.55 + 0.60 + 0.60 + 0.65 + 0.70 + 0.75}{10}.$$

The sum is 4.50, so $\bar{p} = 0.45$. Since $0.45 < 0.5$, the average–probability rule classifies the observation as **Green**.

So:

- Majority vote: Red.
- Average probability: Green.

6. Provide a detailed explanation of the algorithm that is used to fit a regression tree.

Solution: We need to explain the algorithm for fitting a *regression* tree in simple terms. A regression tree predicts a numeric response Y by partitioning the feature space into rectangular regions and predicting the mean Y within each region. The algorithm has two main stages: *growing* a large tree by recursive binary splitting, and then *pruning* it by cost–complexity pruning with cross–validation.

Growing the tree (recursive binary splitting).

1. Start with all training observations in a single region.
2. At each step, consider all predictors $X_1, \ldots, X_p$ and all possible cutpoints s for each predictor.
3. For each candidate split $X_j < s$ versus $X_j \geq s$, compute the sum of squared residuals (RSS) within the two resulting regions:

$$\sum_{i:x_i \in R_1(j,s)} (y_i - \bar{y}_{R_1})^2 + \sum_{i:x_i \in R_2(j,s)} (y_i - \bar{y}_{R_2})^2,$$

   where $\bar{y}_{R_1}$ and $\bar{y}_{R_2}$ are the mean responses in each region .
4. Choose the split (j, s) that gives the largest decrease in RSS (equivalently, the smallest value of the expression above).
5. Apply that split to the appropriate region, creating two child regions.
6. Repeat steps 2–5, always splitting one existing region at a time, until a stopping rule is met (for example, no region has more than a specified minimum number of observations) .

At the end of this stage, we have a large tree that usually overfits the training data.

Pruning the tree (cost–complexity pruning).

1. Given the large tree T_0, consider subtrees T obtained by pruning (cutting off) lower branches.
2. For a tuning parameter $\alpha \geq 0$, define the cost–complexity criterion

$$\sum_{m \in T} \sum_{i:x_i \in R_m} (y_i - \bar{y}_{R_m})^2 + \alpha\,|T|,$$

   where $|T|$ is the number of terminal nodes (leaves) .
3. For each value of α, find the subtree T_α of T_0 that minimizes this criterion; larger α favors smaller trees.
4. Use cross–validation to choose the best α: for each fold,
   - grow a large tree on the training folds,

- compute the sequence of T_α trees,
- measure mean squared prediction error on the validation fold for each α.

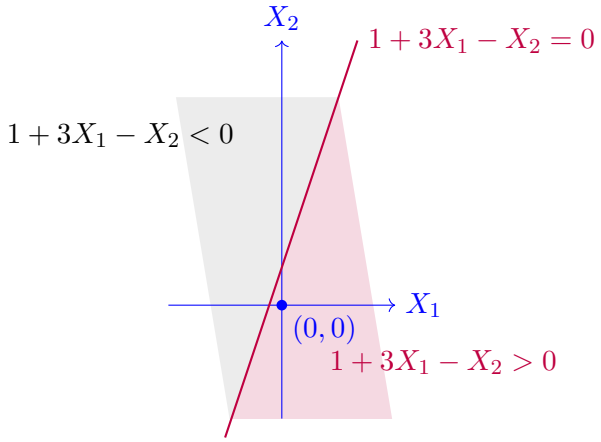
5. Average the validation errors over folds, pick the α with smallest average error, and then refit on the full data to get the final pruned tree corresponding to that α .

Prediction for a new observation x is then simply the mean of Y in the terminal node (region) R_m that x falls into .

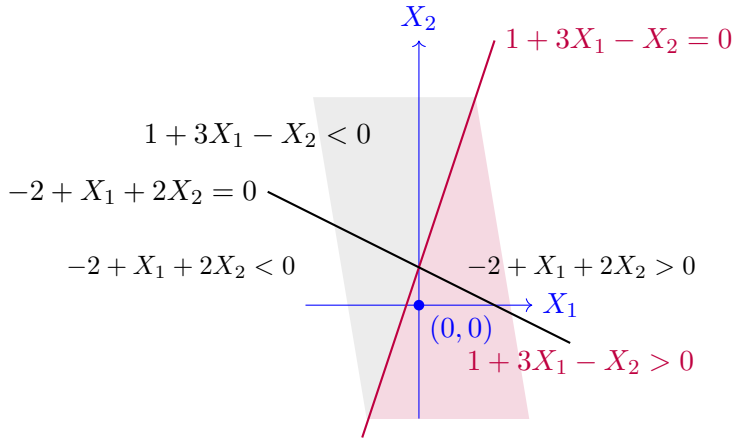
Chapter 9: Support Vector Machines

1. This problem involves hyperplanes in two dimensions.
   (a) Sketch the hyperplane $1 + 3X_1 - X_2 = 0$. Indicate the set of points for which $1 + 3X_1 - X_2 > 0$, as well as the set of points for which $1 + 3X_1 - X_2 < 0$.
   (b) On the same plot, sketch the hyperplane $-2 + X_1 + 2X_2 = 0$. Indicate the set of points for which $-2 + X_1 + 2X_2 > 0$, as well as the set of points for which $-2 + X_1 + 2X_2 < 0$.

Solution: The said hyperplane are drawn below:



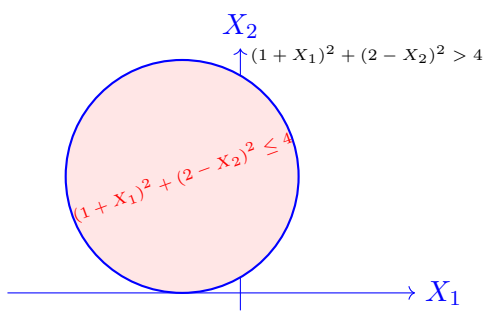
(a) First hyperplane



(b) Both hyperplanes plotted together

2. We have seen that in $p = 2$ dimensions, a linear decision boundary takes the form $\beta_0 + \beta_1 X_1 + \beta_2 X_2 = 0$. We now investigate a non-linear decision boundary.
   (a) Sketch the curve
   $$(1 + X_1)^2 + (2 - X_2)^2 = 4.$$
   (b) On your sketch, indicate the set of points for which
   $$(1 + X_1)^2 + (2 - X_2)^2 > 4,$$
   as well as the set of points for which
   $$(1 + X_1)^2 + (2 - X_2)^2 \leq 4.$$
   (c) Suppose that a classifier assigns an observation to the blue class if
   $$(1 + X_1)^2 + (2 - X_2)^2 > 4,$$
   and to the red class otherwise. To what class is the observation $(0, 0)$ classified? $(-1, 1)$? $(2, 2)$? $(3, 8)$?
   (d) Argue that while the decision boundary in (c) is not linear in terms of X_1 and X_2, it is linear in terms of X_1, X_1^2, X_2, and X_2^2.

Solution: Figures for part (a) and part (b) are as follows:



(c) The classification rule is based on evaluating $V = (1 + X_1)^2 + (2 - X_2)^2$.

Observation	$V(X_1, X_2)$	$V \leq 4$?	**Class**
$(0, 0)$	$(1+0)^2 + (2-0)^2 = 1 + 4 = 5$	Yes	**Blue**
$(-1, 1)$	$(1-1)^2 + (2-1)^2 = 0 + 1 = 1$	No	**Red**
$(2, 2)$	$(1+2)^2 + (2-2)^2 = 9 + 0 = 9$	Yes	**Blue**
$(3, 8)$	$(1+3)^2 + (2-8)^2 = 16 + 36 = 52$	Yes	**Blue**

(d) Even though the decision boundary is not linear in (X_1, X_2), the rule is "blue if

$$1 - X_1^2 - X_2^2 > 4,$$

red otherwise."

Define new features

$$Z_1 = X_1, \quad Z_2 = X_2, \quad Z_3 = X_1^2, \quad Z_4 = X_2^2.$$

Then the inequality becomes

$$1 - Z_3 - Z_4 > 4 \quad \Leftrightarrow \quad -Z_3 - Z_4 - 3 > 0,$$

which is a linear inequality in (Z_1, Z_2, Z_3, Z_4).

So the boundary is linear in the enlarged four-dimensional feature space even though it is nonlinear in the original two-dimensional space. This illustrates the general idea used by support vector machines with polynomial or other nonlinear kernels.

3. Here we explore the maximal margin classifier on a toy data set.

(a) We are given $n = 7$ observations in $p = 2$ dimensions. For each observation, there is an associated class label.

Obs.	X_1	X_2	Y
1	3	4	Red
2	2	2	Red
3	4	4	Red
4	1	4	Red
5	2	1	Blue
6	4	3	Blue
7	4	1	Blue

Sketch the observations.

(b) Sketch the optimal separating hyperplane, and provide the equation for this hyperplane (of the form (9.1)).

(c) Describe the classification rule for the maximal margin classifier. It should be something along the lines of "Classify to Red if $\beta_0 + \beta_1 X_1 + \beta_2 X_2 > 0$, and classify to Blue otherwise." Provide the values for β_0, β_1, and β_2.

(d) On your sketch, indicate the margin for the maximal margin hyperplane.

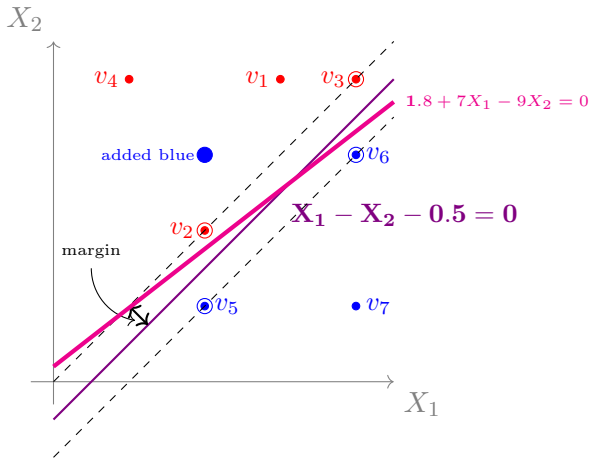
(e) Indicate the support vectors for the maximal margin classifier.

(f) Argue that a slight movement of the seventh observation would not affect the maximal margin hyperplane.

(g) Sketch a hyperplane that is *not* the optimal separating hyperplane, and provide the equation for this hyperplane.

(h) Draw an additional observation on the plot so that the two classes are no longer separable by a hyperplane.

Solution: The following diagram will suffice for parts (a) to (b):



(a) In the above diagram, the observations in "RED" class are shown as "red dots" and the observations in "BLUE" class are shown as "blue dots".

(b) Any line of the for $\beta_0 + \beta_1 X_1 + \beta_2 X_2 = 0$, that is (i) separating and (ii) has the largest possible margin relative to your sketch is the maximal margin hyperplane.
The optimal hyperplane is given by the equation

$$X_1 - X_2 - 0.5 = 0, \qquad \text{or,} \qquad 2X_1 - 2X_2 - 1 = 0,$$

which is shown as the straight line in the graph.

(c) **Classification Rule:** "Classify to Red if $1 - 2X_1 + 2X_2 < 0$, and classify to Blue otherwise."

(d) The margin for the maximal margin hyperplane is 0.5.

(e) Support vectors are the training points that lie exactly on the dashed margin lines, or that would violate the margin in the soft-margin case. In this diagram $(2, 2), (4, 4), (2, 1)$ and $(4, 3)$ are the support vectors.

(f) A slight movement of observation 7 that keeps it on the same side of the margin will not change the maximal margin hyperplane, because only the support vectors control the position of the hyperplane.

(g) See figure for the hyperplane $1.8 + 7X_1 - 9X_2 = 0$, which is separating but not optimal as the margin is < 0.5.

(h) To make the classes non-separable by any straight line, add a new observation so that no line can separate all reds from all blues correctly. For example, let us add a new "BLUE" class observation at $(2, 3)$.

Chapter 10: Deep Learning

1. Consider a neural network with two hidden layers: $p = 4$ input units, 2 units in the first hidden layer, 3 units in the second hidden layer, and a single output.

   (a) Draw a picture of the network, similar to Figures 10.1 or 10.4.

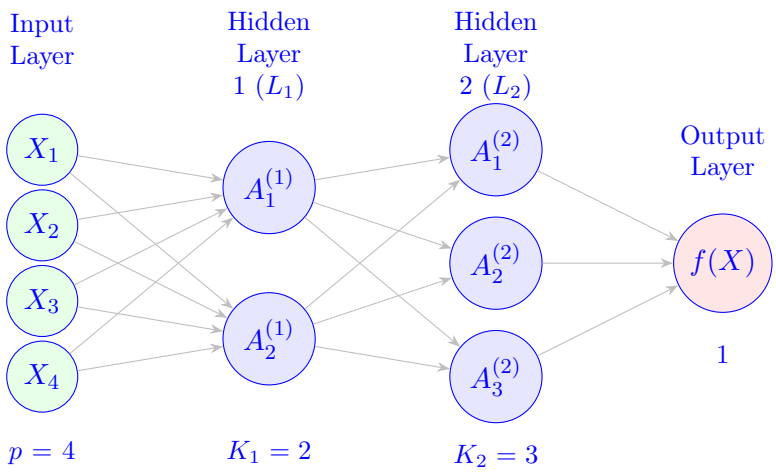
   (b) Write out an expression for $f(X)$, assuming ReLU activation functions. Be as explicit as you can!

   (c) Now plug in some values for the coefficients and write out the value of $f(X)$.

   (d) How many parameters are there?

Solution:

(a) **Network Diagram:**



(b) The ReLU activation function is $g(z) = (z)_+ = \max(0, z)$. The expression for $f(X)$ follows the chain of transformations:

Layer 1 Activations ($A^{(1)}$):

$$A_k^{(1)} = \left(w_{k0}^{(1)} + \sum_{j=1}^{4} w_{kj}^{(1)} X_j \right)_+ \quad \text{for } k = 1, 2$$

Layer 2 Activations ($A^{(2)}$):

$$A_\ell^{(2)} = \left(w_{\ell 0}^{(2)} + \sum_{k=1}^{2} w_{\ell k}^{(2)} A_k^{(1)} \right)_+ \quad \text{for } \ell = 1, 2, 3$$

Output Layer ($f(X)$): Assuming a quantitative response (linear output):

$$f(X) = \beta_0 + \sum_{\ell=1}^{3} \beta_\ell A_\ell^{(2)}$$

Where $A_k^{(1)}$ and $A_\ell^{(2)}$ are implicitly functions of X.

(c) Let input $\mathbf{X} = (1, 0, 0, 0)$.

   (a) **Assume L1 Weights:** Let $w_{k0}^{(1)} = 1$ and $w_{kj}^{(1)} = 1$ for all k, j.

$$A_k^{(1)} = (1 + w_{k1}^{(1)}(1) + 0 + 0 + 0)_+ = (1 + 1)_+ = 2$$

   Thus, $A_1^{(1)} = 2$ and $A_2^{(1)} = 2$.

(b) **Assume L2 Weights:** Let $w_{\ell 0}^{(2)} = 0.5$ and $w_{\ell k}^{(2)} = 0.5$ for all ℓ, k.

$$A_\ell^{(2)} = (0.5 + 0.5A_1^{(1)} + 0.5A_2^{(1)})_+ = (0.5 + 0.5(2) + 0.5(2))_+ = (0.5 + 1 + 1)_+ = 2.5$$

Thus, $A_1^{(2)} = 2.5, A_2^{(2)} = 2.5, A_3^{(2)} = 2.5$.

(c) **Assume Output Weights:** Let $\beta_0 = 0$ and $\beta_\ell = 1$ for $\ell = 1, 2, 3$.

$$f(X) = 0 + \sum_{\ell=1}^{3} (1)A_\ell^{(2)} = 0 + 2.5 + 2.5 + 2.5 = \mathbf{7.5}$$

(d) The total number of parameters (weights w and biases β_0) is calculated layer by layer:

- **Input (4) to L1 (2):** 2 units $\times$ (4 inputs + 1 bias) = 10 parameters.
- **L1 (2) to L2 (3):** 3 units $\times$ (2 inputs + 1 bias) = 9 parameters.
- **L2 (3) to Output (1):** 1 unit $\times$ (3 inputs + 1 bias β_0) = 4 parameters.

$$\text{Total Parameters} = 10 + 9 + 4 = \mathbf{23}$$

2. Consider the *softmax* function in (10.13) (see also (4.13) on page 141) for modeling multinomial probabilities.

(a) In (10.13), show that if we add a constant c to each of the z_m, $m = 1, \ldots, K$, then the probabilities in (10.13) remain unchanged. (*Hint*: $e^{a+c} = e^a e^c$.)

(b) In (4.13), show that if we add constants $c_1, \ldots, c_K$ to each of the numerators and denominators, then the probabilities remain unchanged if $c_1 = \cdots = c_K$.

Solution:

a) Let $Z'_\ell = Z_\ell + c$. The new estimated probability $f'_m(X)$ is:

$$f'_m(X) = \frac{e^{Z'_m}}{\sum_{\ell=0}^{9} e^{Z'_\ell}} = \frac{e^{Z_m + c}}{\sum_{\ell=0}^{9} e^{Z_\ell + c}}$$

Using the property $e^{A+B} = e^A e^B$:

$$f'_m(X) = \frac{e^c e^{Z_m}}{\sum_{\ell=0}^{9} e^c e^{Z_\ell}} = \frac{e^c e^{Z_m}}{e^c \sum_{\ell=0}^{9} e^{Z_\ell}}$$

Since e^c appears as a common factor in both the numerator and the denominator, they cancel out:

$$f'_m(X) = \frac{e^{Z_m}}{\sum_{\ell=0}^{9} e^{Z_\ell}} = f_m(X)$$

Thus, adding a constant c to all linear predictors Z_ℓ leaves the softmax probability unchanged.

b) In multinomial logistic regression, the linear predictor for class m is $Z_m = \beta_{m0} + \sum_{j=1}^{p} \beta_{mj} X_j$. Using $X_0 = 1$ for the intercept, we can write $Z_m = \sum_{j=0}^{p} \beta_{mj} X_j$.

We define new coefficients $\beta'_{mj} = \beta_{mj} + c_j$, where c_j are constants added to the corresponding coefficient (including c_0 added to the intercept β_{m0}) for *each* class m.

The new linear predictor Z'_m is:

$$Z'_m = \sum_{j=0}^{p} (\beta_{mj} + c_j) X_j = \sum_{j=0}^{p} \beta_{mj} X_j + \sum_{j=0}^{p} c_j X_j$$

$$\implies Z'_m = Z_m + C, \qquad \text{where} \quad C = \sum_{j=0}^{p} c_j X_j.$$

Since the input vector X is fixed for a given test point x, C is a constant value that does **not** depend on the class index m. By the result proven in part (a), adding a class-independent constant C to every Z_m does not change the resulting softmax probability $f_m(X)$. This demonstrates that the softmax function is over-parametrized.

3. Show that the negative multinomial log-likelihood (10.14) is equivalent to the *negative log* of the likelihood expression (4.5) when there are $M = 2$ classes.

Solution: The negative multinomial log-likelihood (or cross-entropy) is defined as:

$$L_{\text{mult}} = -\sum_{i=1}^{n} \sum_{m=0}^{M-1} y_{im} \log(f_m(x_i)) \tag{10.14}$$

For $M = 2$ classes, we use $m = 0$ and $m = 1$ (assuming class indices start at 0).

$$L_{\text{mult}} = -\sum_{i=1}^{n} [y_{i0} \log(f_0(x_i)) + y_{i1} \log(f_1(x_i))] \tag{4.5}$$

In the case of two classes (binary classification), the dummy variable representation requires $y_{i0} + y_{i1} = 1$. Thus, $y_{i0} = 1 - y_{i1}$. Similarly, the estimated probabilities must sum to one: $f_0(x_i) = 1 - f_1(x_i)$.

Substituting these binary relationships into the likelihood expression:

$$L_{\text{mult}} = -\sum_{i=1}^{n} [(1 - y_{i1}) \log(1 - f_1(x_i)) + y_{i1} \log(f_1(x_i))]$$

This expression is the negative log-likelihood used for two-class logistic regression, which is the generalization of criterion (4.5) mentioned in the text.

4. Consider a CNN that takes in 32×32 grayscale images and has a single convolution layer with three 5×5 convolution filters (without boundary padding).

   (a) Draw a sketch of the input and first hidden layer similar to Figure 10.8.

   (b) How many parameters are in this model?

   (c) Explain how this model can be thought of as an ordinary feed-forward neural network with the individual pixels as inputs, and with constraints on the weights in the hidden units. What are those constraints?

   (d) If there were no constraints, then how many weights would there be in the ordinary feed-forward neural network in (c)?

Solution:

(a)

(b) The model has $K = 3$ convolution filters. Each filter is 5×5 and operates on the 1 input channel (grayscale).

- Weights per filter: $5 \times 5 = 25$ weights.
- Biases per filter: 1 bias term.
- Total parameters: $3 \times (25 + 1) = \mathbf{78}$ parameters.

(c) An ordinary feed-forward network would treat the $32 \times 32 = 1024$ pixels as $p = 1024$ inputs. The convolution layer produces $3 \times 28 \times 28 = 2352$ hidden units.

This convolutional layer is equivalent to a fully connected feed-forward layer between 1024 inputs and 2352 hidden units, subject to two main constraints that drastically reduce the number of free parameters:

(a) **Local Connectivity (Sparsity):** Each hidden unit (one output pixel) is only connected to a small, localized region (the 5×5 patch) of the input layer. The vast majority of potential weights connecting the input layer to the hidden layer are structurally set to zero.

(b) **Weight Sharing:** The same set of 25 weights (plus one bias) that define a specific convolution filter are reused across all 28×28 positions in that output feature map. This reuse (or weight sharing) is what makes the model computationally efficient and parameter-sparse.

(d) Without constraints, the layer would be fully connected:

- Input Units (p): $32 \times 32 = 1024$.
- Hidden Units (K): $3 \times 28 \times 28 = 2352$.
- Total weights (Input-to-Hidden): $1024 \times 2352 = 2,408,448$.
- Total biases (Hidden Unit Biases): 2352.

$$\text{Total Parameters} = 2,408,448 + 2,352 = \mathbf{2,410,800}$$

5. In Table 10.2 on page 433 of the book, we see that the ordering of the three methods with respect to mean absolute error is different from the ordering with respect to test set R^2. How can this be?

Solution: The difference in ranking between the three methods (Linear Regression, Lasso, Neural Network) is due to the fundamental differences in how Mean Absolute Error (MAE) and Test Set R^2 measure prediction error.

- **Mean Absolute Error (MAE):** MAE uses the absolute value of the residuals (L1 loss). It is robust to outliers and represents the average magnitude of the error.
- **Test Set R^2:** R^2 is based on the squared residuals (L2 loss). Because errors are squared, this metric disproportionately penalizes large errors or outliers.

Note: *A model (like the Lasso in the given table) might achieve a low MAE because it produces many small errors, but its R^2 might suffer if it commits even a few large errors, as these outliers are heavily weighted (magnified) by the squaring operation inherent in the R^2 calculation. Conversely, a model that performs slightly worse on MAE might perform better on R^2 if it is successful in minimizing the incidence of these catastrophic, large outliers, resulting in different orderings depending on the chosen metric.*